35

The Nature of Light and the Principles of Ray Optics

CHAPTER OUTLINE

35.1 The Nature of Light

35.2 Measurements of the Speed of Light

35.3 The Ray Approximation in Ray Optics

35.4 Analysis Model: Wave Under Reflection

35.5 Analysis Model: Wave Under Refraction

35.6 Huygens's Principle

35.7 Dispersion

35.8 Total Internal Reflection

* An asterisk indicates a question or problem new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ35.1 The ranking is answer e, c, b, a, d. We consider the quantity λ/d : the smaller it is, the better the ray approximation works. The quantity λ/d is about (a) 0.34 m/1 m $\approx$ 0.3, (b) 0.7 μm/2 mm $\approx$ 0.000 3, (c) 0.4 μm/2 mm $\approx$ 0.000 2, (d) 300 m/1 m $\approx$ 300, (e) 1 nm/1 mm $\approx$ 0.000 001.

OQ35.2 Answer (c). As light travels from one medium to another, both the wavelength of the light and the index of refraction of the medium will change, but the product λn is constant: $\lambda_2 n_2 = \lambda_{\text{air}} n_{\text{air}}$. In going from air into a second medium of index n, according to Equation 25.6, $n = \lambda/\lambda_n = 495 \text{ nm}/434 \text{ nm} = 1.14$.

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OQ35.3 Answer (b). In going from carbon disulfide ($n_1 = 1.63$) to crown glass ($n_2 = 1.52$), the critical angle for total internal reflection is

$$\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right) = \sin^{-1}\left(\frac{1.52}{1.63}\right) = 68.8°$$

OQ35.4 Answers (a), (b), and (c) are all correct statements. The frequency of a wave does not change when it travels from one medium to another: $f_1 = f_2 \rightarrow n_1\lambda_1 = n_2\lambda_2$; also, Snell's law of refraction states $n_1 \sin\theta_1 = n_2 \sin\theta_2$. By their definitions, $n = c/v = c/f\lambda$ and $\sin\theta = 1/\csc\theta$. Thus, Snell's law can take these alternate forms:

$$\frac{\sin\theta_1}{v_1} = \frac{\sin\theta_2}{v_2} \rightarrow \frac{v_1}{\sin\theta_1} = \frac{v_2}{\sin\theta_2} \rightarrow \frac{\csc\theta_1}{n_1} = \frac{\csc\theta_2}{n_2} \rightarrow \frac{\lambda_1}{\sin\theta_1} = \frac{\lambda_2}{\sin\theta_2}$$

Snell originally stated his law in terms of cosecants.

OQ35.5 Answer (e). The index of refraction of glass is greater than that of air, which means the speed of light in glass is slower than in air ($n = c/v$). The frequency does not change, but because the speed decreases, the wavelength also decreases.

OQ35.6 Answer (b). When light is in water, the relationships between the values of its speed and wavelength to the values of the same quantities in air are $n_{\text{water}} = \dfrac{c}{v_{\text{water}}} \rightarrow v_{\text{water}} = \dfrac{c}{n_{\text{water}}} = \dfrac{3}{4}c$, and

$$n_{\text{water}}\lambda_{\text{water}} = n_{\text{air}}\lambda_{\text{air}} \rightarrow \lambda_{\text{water}} = \left(\frac{n_{\text{air}}}{n_{\text{water}}}\right)\lambda_{\text{air}} \approx \frac{3}{4}\lambda_{\text{air}}.$$

OQ35.7 Answer (c). Water has a greater index of refraction than air. In passing from one of these media into the other, light will be refracted (deviated in direction) unless the angle of incidence is zero (in which case, the angle of refraction is also zero). Because the angle of refraction can be zero only if the angle of incidence is zero, ray *B* cannot be correct. In refraction, the incident ray and the refracted ray are never on the same side of the line normal to the surface at the point of contact, so ray *A* cannot be correct. Also in refraction, $n_2 \sin\theta_2 = n_1 \sin\theta_1$; thus, if $n_2 > n_1$, then $\theta_2 < \theta_1$: the refracted ray makes a smaller angle with the normal in the medium having the higher index of refraction. Therefore, rays *D* and *E* cannot be correct, leaving only ray *C* as a likely path.

OQ35.8 Answer (c). The time interval is $10^4 \text{ m}/(3 \times 10^8 \text{ m/s}) = 33\ \mu s$.

OQ35.9 Answer (c). For any medium, other than vacuum, the index of refraction for red light is slightly lower (closer to 1) than that for blue light. This means that when light goes from vacuum (or air) into

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glass, the red light deviates from its original direction less than does the blue light. Also, as the light reemerges from the glass into vacuum (or air), the red light again deviates less than the blue light. If the two surfaces of the glass are parallel to each other, the red and blue rays will emerge traveling parallel to each other, but displaced laterally from one another. The sketch that best illustrates this process is *C*.

OQ35.10 For a wave to experience total internal reflection, it must be traveling in the medium in which it moves slower, in which it has a greater index of refraction.

(i) Answer (a). Water has a greater index of refraction than air.

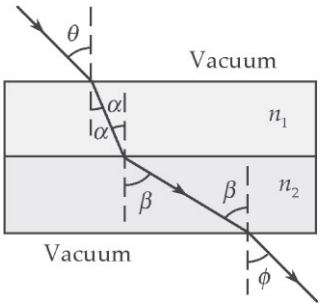
(ii) Answer (c). The sound travels slower in air than in water.

OQ35.11 Answer (c). Consider the sketch in ANS. FIG. OQ35.11 and apply Snell's law to the refraction at each of the three surfaces. Because the surfaces are parallel, the resulting equations are

$(1.00)\sin\theta = n_1 \sin\alpha$ (Top surface)

$n_1 \sin\alpha = n_2 \sin\beta$ (Middle surface)

$n_2 \sin\beta = (1.00)\sin\phi$ (Bottom surface)



ANS. FIG. OQ35.11

These equations allow us to equate the left side of the first equation with the right side of the last equation:

$$(1.00)\sin\theta = (1.00)\sin\phi \rightarrow \phi = \theta$$

OQ35.12 Color A travels slower in the glass of the prism. Light with the greater change in speed will have the greater deviation in direction.

OQ35.13 Answer (c). We want a big difference between indices of refraction to have total internal reflection under the widest range of conditions.

OQ35.14 Answer (a). In a dispersive medium, the index of refraction is largest for the shortest wavelength. Thus, the violet light will be refracted (or bent) the most as it passes through a surface of the crown glass.

OQ35.15 Answer (b). For a wave to experience total internal reflection, it must be traveling in the medium in which it moves slower, in which it has a greater index of refraction. A light ray, in attempting to go from a medium with index of refraction n_1 into a second medium with index of refraction n_2, will undergo total internal reflection if $n_2 < n_1$ and if the ray strikes the surface at an angle of incidence greater than or equal to the critical angle.

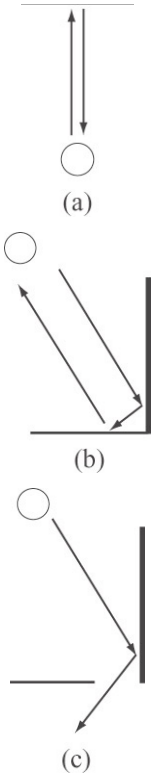
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ANSWERS TO CONCEPTUAL QUESTIONS

CQ35.1 The water level in a clear glass is observable because light is refracted as it passes from air to water to air. The index of liquid helium is very close to that of air, so very little refraction occurs as light travels from air to helium to the air.

CQ35.2 At the altitude of the plane the surface of the Earth need not block off the lower half of the rainbow. Thus, the full circle can be seen. You can see such a rainbow by climbing on a stepladder above a garden sprinkler in the middle of a sunny day. Set the sprinkler for fine mist. Do not let the slippery children fall from the ladder.

CQ35.3 (a) We assume that you and the child are always standing close together. For a flat wall to make an echo of a sound that you make, you must be standing along a normal to the wall. You must be on the order of 100 m away, to make the transit time sufficiently long that you can hear the echo separately from the original sound. Your sound must be loud enough so that you can hear it even at this considerable range. In ANS. FIG. CQ35.3(a), the circle represents an area in which you can be standing. The arrows represent rays of sound.



ANS FIG. CQ35.3

(b) Now suppose two vertical perpendicular walls form an inside corner that you can see. Some of the sound you radiate horizontally will be headed generally toward the corner. It will reflect from both walls with high efficiency to reverse in direction and come back to you, as shown in ANS. FIG. CQ35.3(b). You can stand anywhere reasonably far away to hear a retroreflected echo of sound you produce.

(c) If the two walls are not perpendicular, the inside corner will not produce retroreflection. You will generally hear no echo of your shout or clap.

(d) If two perpendicular walls have a reasonably narrow gap between them at the corner, you can still hear a clear echo. It is not the corner line itself that retroreflects the sound, but the perpendicular walls on both sides of the corner. [ANS. FIG. CQ35.3(b) applies also in this case.]

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(e) At some angles, sound will reflect from the first wall but not the second; rather, it will pass into the breezeway, as shown in ANS. FIG. CQ35.3(c), so there will be no echo.

CQ35.4 The stealth fighter is designed so that adjacent panels are not joined at right angles, to prevent any retroreflection of radar signals. This means that radar signals directed at the fighter will not be channeled back toward the detector by reflection. Just as with sound, radar signals can be treated as *diverging* rays, so that any ray that is by chance reflected back to the detector will be too weak in intensity to distinguish from background noise.

CQ35.5 "Immediately around the dark shadow of my head, I see a halo brighter than the rest of the dewy grass." It is called the *heiligenschein*. Cellini believed that it was a miraculous sign of divine favor pertaining to him alone. Apparently none of the people to whom he showed it told him that they could see halos around their own shadows but not around Cellini's. Thoreau knew that each person had his own halo. He did not draw any ray diagrams but assumed that it was entirely natural. Between Cellini's time and Thoreau's, the Enlightenment and Newton's explanation of the rainbow had happened. Today the effect is easy to see whenever your shadow falls on a retroreflecting traffic sign, license plate, or road stripe. When a bicyclist's shadow falls on a paint stripe marking the edge of the road, her halo races along with her.

CQ35.6 An echo is an example of the reflection of sound. Hearing the noise of a distant highway on a cold morning, when you cannot hear it after the ground warms up, is an example of acoustical refraction. You can use a rubber inner tube (or balloon of the same shape) inflated with helium as an acoustical lens to concentrate sound in the way a lens can focus light: the speed of sound is greater in helium, so wavefronts passing through the helium speed ahead of wavefronts passing through the air in the doughnut hole of the tube, so that the overall shape of the wavefronts changes from plane to concave, resulting in a focusing of the wave. At your next party, see if you can experimentally find the approximate focal point!

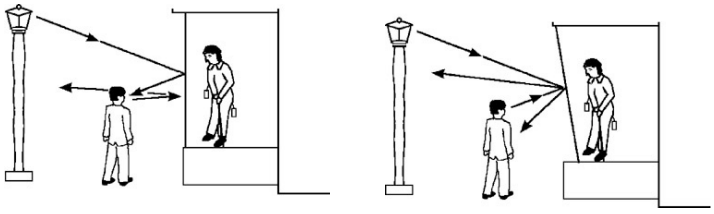
CQ35.7 Highly silvered mirrors reflect about 98% of the incident light. With a 2-mirror periscope, that results in approximately a 4% decrease in intensity of light as the light passes through the periscope. This may not seem like much, but in low-light conditions, that lost light may mean the difference between being able to distinguish an enemy armada or an iceberg from the sky beyond. Using prisms results in total internal reflection, meaning that 100% of the incident light is reflected through the periscope. That is the "total" in total internal reflection.

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CQ35.8 Diamond has higher index of refraction than glass and consequently a smaller critical angle for total internal reflection. A brilliant-cut diamond is shaped to admit light from above, reflect it totally at the converging facets on the underside of the jewel, and let the light escape only at the top. Glass will have less light internally reflected.

CQ35.9 If a laser beam enters a sugar solution with a concentration gradient (density and index of refraction increasing with depth), then the laser beam will be progressively bent downward (toward the normal) as it passes into regions of greater index of refraction.

CQ35.10 With a vertical shop window, streetlights and his own reflection can impede the window shopper's clear view of the display. The tilted shop window can put these reflections out of the way. Windows of airport control towers are also tilted like this, as are automobile windshields.



ANS. FIG. CQ35.10

CQ35.11 (a) Light from the lamps along the edges of the sheet enters the plastic, and then the front and back faces of the plastic totally internally reflect it, wherever the plastic has an interface with air. If the refractive index of the grease is intermediate between 1.55 and 1.00, some of this light can leave the plastic into the grease and leave the grease into the air. The surface of the grease is rough, so the grease can send out light in all directions. The customer sees the grease shining against a black background.

(b) The spotlight method of producing the same effect is much less efficient. With it, the blackboard absorbs much of the light from the spotlight.

(c) The refractive index of the grease must be less than 1.55. Perhaps the best choice would be $\sqrt{1.55 \times 1.00} = 1.24$.

CQ35.12 A mirage occurs when light changes direction as it moves between batches of air having different indices of refraction because they have different densities at different temperatures. When the sun makes a blacktop road hot, an apparent wet spot is bright due to refraction of light from the bright sky. The light, originally headed a little below the horizontal, always bends up as it first enters and then leaves sequentially hotter, lower-density, lower-index layers of air closer to the road surface.

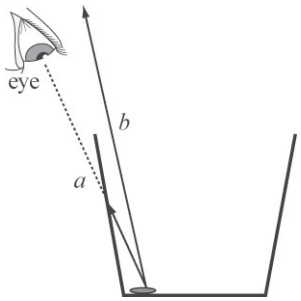
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CQ35.13 Light rays coming from parts of the pencil under water are bent away from the normal as they emerge into the air above. The rays enter the eye (or camera) at angles closer to the horizontal, thus the parts of the pencil under water appear closer to the surface than they actually are, so the pencil appears bent. See CQ35.16 for an illustration of a related effect.

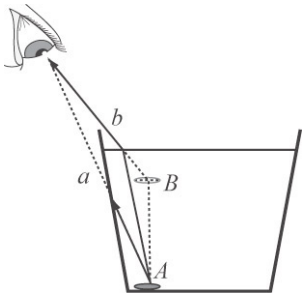
CQ35.14 No. The speed of light v in any medium except vacuum is less than the speed of light c in vacuum. By definition, the index of refraction $n = c/v$, thus the index of any material medium is always greater than 1. A material with an index less than 1 is impossible.

CQ35.15 Light travels through a vacuum at a speed of 300 000 km per second. Thus, an image we see from a distant star or galaxy must have been generated some time ago. For example, the star Altair is 16 light-years away; if we look at an image of Altair today, we know only what was happening 16 years ago. This may not initially seem significant, but astronomers who look at other galaxies can gain an idea of what galaxies looked like when they were significantly younger. Thus, it actually makes sense to speak of "looking backward in time."

CQ35.16 With no water in the cup, light rays from the coin do not reach the eye because they are blocked by the side of the cup. With water in the cup, light rays are bent away from the normal as they leave the water so that some reach the eye.



ANS. FIG. CQ35.16(a)



ANS. FIG. CQ35.16(b)

In ANS. FIG. CQ35.16(a), ray *a* is blocked by the side of the cup so it cannot enter the eye, and ray *b* misses the eye. In ANS. FIG. CQ35.16(b), ray *a* is still blocked by the side of the cup, but ray *b* refracts at the water's surface so that it reaches the eye. Ray *b* seems to come from position *B*, directly above the coin at position *A*.

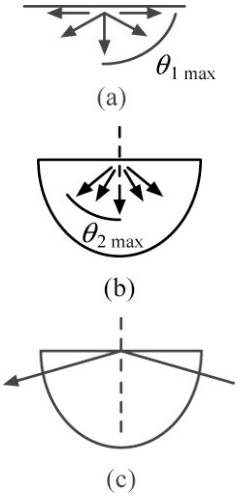
CQ35.17 (a) Scattered light rays leave the center of the photograph, shown in ANS. FIG. CQ35.17(a), in all horizontal directions between $\theta_1 = 0°$ and 90° from the normal. When the light rays immediately enter the

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water they are gathered into a fan, shown in ANS. FIG. CQ35.17(b), between 0° and $\theta_{2\max}$ given by

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$
$$1.00 \sin 90 = 1.333 \sin\theta_{2\ \max}$$
$$\theta_{2\ \max} = 48.6°$$

The light rays leave the cylinder without deviation because they travel along the normal everywhere they strike the surface of the glass, so the viewer only receives light from the center of the photograph when he has turned by an angle less than 48.6°.



ANS. FIG. CQ35.17

(b) When the paperweight is turned farther, light at the back surface undergoes total internal reflection, shown in ANS. FIG. CQ35.17(c). The viewer sees things outside the globe on the far side.

SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 35.1 The Nature of Light

Section 35.2 Measurements of the Speed of Light

***P35.1** We find the energy of the photons from Equation 35.1, $E = hf$.

(a) $$E = hf = \left(6.63\times10^{-34}\ \text{J}\cdot\text{s}\right)\left(5.00\times10^{17}\ \text{Hz}\right)\left(\frac{1\ \text{eV}}{1.60\times10^{-19}\ \text{J}}\right)$$

$$= \boxed{2.07\times10^{3}\ \text{eV}} = 2.07\ \text{keV}$$

(b) $$E = hf = \frac{hc}{\lambda}$$

$$= \frac{\left(6.63\times10^{-34}\ \text{J}\cdot\text{s}\right)\left(3.00\times10^{8}\ \text{m/s}\right)}{3.00\times10^{2}\ \text{nm}}\left(\frac{1\ \text{nm}}{10^{-9}\ \text{m}}\right)\left(\frac{1\ \text{eV}}{1.60\times10^{-19}\ \text{J}}\right)$$

$$= \boxed{4.14\ \text{eV}}$$

P35.2 (a) The Moon's radius is 1.74×10^6 m and the Earth's radius is 6.37×10^6 m. The total distance traveled by the light is:

$$d = 2\left(3.84\times10^{8}\ \text{m} - 1.74\times10^{6}\ \text{m} - 6.37\times10^{6}\ \text{m}\right)$$
$$= 7.52\times10^{8}\ \text{m}$$

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This takes 2.51 s, so

$$v = \frac{7.52 \times 10^8 \text{ m}}{2.51 \text{ s}} = \boxed{3.00 \times 10^8 \text{ m/s}}$$

(b) The sizes of the objects need to be taken into account. Otherwise the answer would be too large by 2%.

P35.3 The experiment is most convincing if the wheel turns fast enough to pass outgoing light through one notch and returning light through the next. This requires $\Delta t = \frac{2\ell}{c}$, or

$$\theta = \omega \Delta t = \omega\left(\frac{2\ell}{c}\right)$$

so $$\omega = \frac{c\theta}{2\ell} = \frac{\left(2.998 \times 10^8 \text{ m/s}\right)\left[2\pi/(720)\right]}{2\left(11.45 \times 10^3 \text{ m}\right)} = \boxed{114 \text{ rad/s}}$$

The returning light would be blocked by a tooth at one-half the angular speed, giving another data point.

P35.4 The difference is due to the extra time light takes to cross Earth's orbit. From $\Delta x = c\Delta t$, we have

$$c = \frac{\Delta x}{\Delta t} = \frac{2\left(1.50 \times 10^8 \text{ km}\right)\left(1000 \text{ m/km}\right)}{(22.0 \text{ min})(60.0 \text{ s/min})} = \boxed{2.27 \times 10^8 \text{ m/s}}$$

Section 35.3 The Ray Approximation in Ray Optics

Section 35.4 Analysis Model: Wave Under Reflection

Section 35.5 Analysis Model: Wave Under Refraction

P35.5 (a) $$f = \frac{c}{\lambda} = \frac{3.00 \times 10^8 \text{ m/s}}{6.328 \times 10^{-7} \text{ m}} = \boxed{4.74 \times 10^{14} \text{ Hz}}$$

(b) $$\lambda_{\text{glass}} = \frac{\lambda_{\text{air}}}{n} = \frac{632.8 \text{ nm}}{1.50} = \boxed{422 \text{ nm}}$$

(c) $$v_{\text{glass}} = \frac{c_{\text{air}}}{n} = \frac{3.00 \times 10^8 \text{ m/s}}{1.50} = \boxed{2.00 \times 10^8 \text{ m/s}}$$

P35.6 Refracted light enters the diver's eyes. The angle of refraction θ_2 is 45.0°. From Snell's law,

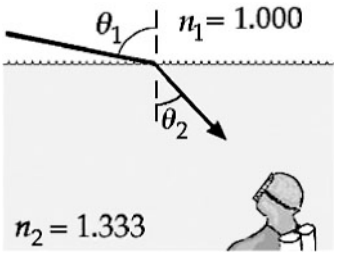
$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

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Solving,

$$\theta_1 = \sin^{-1}(1.333 \sin 45.0°)$$

$$= 70.5° \text{ from the vertical} \rightarrow \boxed{19.5° \text{ above the horizon}}$$



ANS. FIG. P35.6

P35.7 We find the angle of incidence from Snell's law, $n_1 \sin\theta_1 = n_2 \sin\theta_2$. Solving,

$$1.333 \sin\theta_1 = 1.52 \sin 19.6° \quad \rightarrow \quad \theta_1 = 22.5°$$

The angle of reflection of the beam in water is then also $\boxed{22.5°}$.

P35.8 (a) The dashed lines are parallel, and alternate interior angles are equal between parallel lines, so the angle of refraction law at the air-oil interface is 20.0°. Applying Snell's law,

$$n_{\text{air}} \sin\theta = n_{\text{oil}} \sin\alpha$$

$$1.00 \sin\theta = 1.48 \sin 20.0°$$

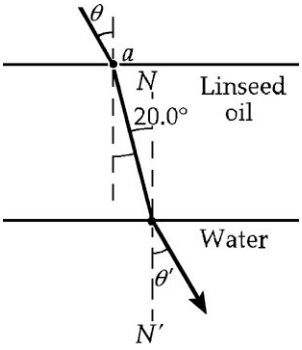
yields $\boxed{\theta = 30.4°}$.

(b) The angle of incidence $\alpha = 20.0°$. Applying Snell's law at the oil-water interface,

$$n_{\text{water}} \sin\theta' = n_{\text{oil}} \sin\alpha$$

$$1.33 \sin\theta' = 1.48 \sin 20.0°$$

yields $\boxed{\theta' = 22.3°}$.



ANS. FIG. P35.8

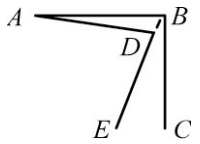
P35.9 (a) flint glass: $v = \dfrac{c}{n} = \dfrac{3.00 \times 10^8 \text{ m/s}}{1.66} = \boxed{1.81 \times 10^8 \text{ m/s}}$

(b) water: $v = \dfrac{c}{n} = \dfrac{3.00 \times 10^8 \text{ m/s}}{1.333} = \boxed{2.25 \times 10^8 \text{ m/s}}$

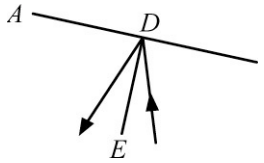
(c) cubic zirconia: $v = \dfrac{c}{n} = \dfrac{3.00 \times 10^8 \text{ m/s}}{2.20} = \boxed{1.36 \times 10^8 \text{ m/s}}$

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P35.10 (a) Let *AB* be the originally horizontal ceiling, *BC* its originally vertical normal, *AD* the new ceiling, and *DE* its normal. Then angle $BAD = \phi$. By definition *DE* is perpendicular to *AD* and *BC* is perpendicular to *AB*. Then the angle between *DE* extended and *BC* is ϕ because angles are equal when their sides are perpendicular, right side to right side and left side to left side.



ANS. FIG. P35.10(a)



ANS. FIG. P35.10(b)

(b) Now $CBE = \phi$ is the angle of incidence of the vertical light beam. Its angle of reflection is also ϕ. The angle between the vertical incident beam and the reflected beam is 2ϕ.

(c) $$\tan 2\phi = \frac{1.40\text{ cm}}{720\text{ cm}} = 0.001\,94 \qquad \boxed{\phi = 0.055\,7^\circ}$$

P35.11 From Snell's law, $n_2 \sin\theta_2 = n_1 \sin\theta_1$. Thus, when $\theta_1 = 45.0^\circ$ and the first medium is air ($n_1 = 1.00$), we have $\sin\theta_2 = (1.00)\sin 45.0^\circ / n_2$.

(a) For quartz, $n_2 = 1.458$:

$$\theta_2 = \sin^{-1}\left(\frac{(1.00)\sin 45.0^\circ}{1.458}\right) = \boxed{29.0^\circ}$$

(b) For carbon disulfide, $n_2 = 1.628$:

$$\theta_2 = \sin^{-1}\left(\frac{(1.00)\sin 45.0^\circ}{1.628}\right) = \boxed{25.7^\circ}$$

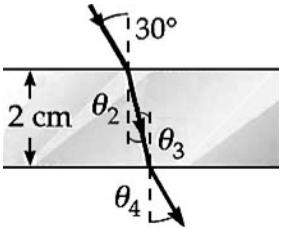
(c) For water, $n_2 = 1.333$:

$$\theta_2 = \sin^{-1}\left(\frac{(1.00)\sin 45.0^\circ}{1.333}\right) = \boxed{32.0^\circ}$$

P35.12 At entry, the wave under refraction model, expressed as $n_1 \sin\theta_1 = n_2 \sin\theta_2$, gives

$$\theta_2 = \sin^{-1}\left(\frac{n_1 \sin\theta_1}{n_2}\right) = \sin^{-1}\left(\frac{1.000\sin 30.0^\circ}{1.50}\right) = \boxed{19.5^\circ}$$

To do ray optics, you must remember some geometry. The surfaces of entry and exit are parallel so their normals are parallel. Then angle θ_2 of refraction at entry and the angle θ_3 of incidence at exit are alternate interior angles formed by the ray as a transversal cutting



ANS. FIG. P35.12

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parallel lines. Therefore, $\theta_3 = \theta_2 = \boxed{19.5°}$.

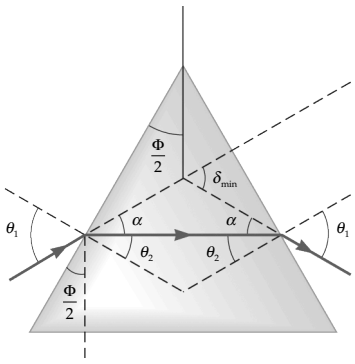
At the exit point, $n_2 \sin\theta_3 = n_1 \sin\theta_4$ gives

$$\theta_4 = \sin^{-1}\left(\frac{n_2 \sin\theta_3}{n_1}\right) = \sin^{-1}\left(\frac{1.50 \sin 19.5°}{1.000}\right) = \boxed{30.0°}$$

Because θ_1 and θ_4 are equal, the departing ray in air is parallel to the original ray.

P35.13 Taking Φ to be the apex angle and $\delta_{\min}$ to be the angle of minimum deviation (See ANS. FIG. P35.13), from Equation 35.9, the index of refraction of the prism material is

$$n = \frac{\sin\left[\left(\Phi + \delta_{\min}\right)/2\right]}{\sin\left(\Phi/2\right)}$$



ANS. FIG. P35.13

Solving for $\delta_{\min}$,

$$\delta_{\min} = 2\sin^{-1}\left(n \sin\frac{\Phi}{2}\right) - \Phi$$

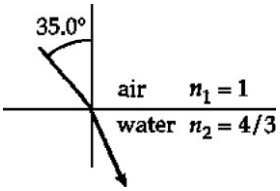
$$= 2\sin^{-1}\left[(2.20)\sin(25.0°)\right] - 50.0°$$

$$= \boxed{86.8°}$$

P35.14 (a) The law of refraction $n_1 \sin\theta_1 = n_2 \sin\theta_2$ can be put into the more general form

$$\frac{c}{v_1}\sin\theta_1 = \frac{c}{v_2}\sin\theta_2$$

$$\frac{\sin\theta_1}{v_1} = \frac{\sin\theta_2}{v_2}$$



ANS. FIG. P35.14

This is equivalent to Equation 35.3. This form applies to all kinds of waves that move through space.

In air at 20°C, the speed of sound is 343 m/s. From Table 17.1, the speed of sound in water at 25.0°C is 1493 m/s. The angle of incidence is 13.0°:

$$\frac{\sin 13.0°}{343 \text{ m/s}} = \frac{\sin\theta_2}{1\,493 \text{ m/s}}$$

$$\theta_2 = \boxed{78.3°}$$

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(b) The wave keeps constant frequency in all media:

$$f = \frac{v_1}{\lambda_1} = \frac{v_2}{\lambda_2}$$

$$\lambda_2 = \frac{v_2 \lambda_1}{v_1} = \frac{1\ 493\ \text{m/s}(0.589\ \text{m})}{343\ \text{m/s}} = \boxed{2.56\ \text{m}}$$

(c) Using Snell's law,

$$n_2 \sin\theta_2 = n_1 \sin\theta_1$$

$$1.333 \sin\theta_2 = 1.000\ 293 \sin 13.0°$$

$$\theta_2 = \boxed{9.72°}$$

(d) $$\lambda_2 = \frac{v_2 \lambda_1}{v_1} = \frac{n_1 \lambda_1}{n_2} = \frac{1.000\ 293(589\ \text{nm})}{1.333} = \boxed{442\ \text{nm}}$$

(e) The light wave slows down as it moves from air to water, but the sound wave speeds up by a large factor. The light wave bends toward the normal and its wavelength shortens, but the sound wave bends away from the normal and its wavelength increases.

***P35.15** From the wave under refraction model, $n_1 \sin\theta_1 = n_2 \sin\theta_2$, we solve for the index of refraction n_2 in the substance:

$$n_2 = \frac{1.333 \sin 37.0°}{\sin 25.0°} = 1.90$$

Then, from the definition of index of refraction,

$$n_2 = 1.90 = \frac{c}{v}: \qquad v = \frac{c}{1.90} = 1.58 \times 10^8\ \text{m/s} = \boxed{158\ \text{Mm/s}}$$

***P35.16** (a) $n_1 \sin\theta_1 = n_2 \sin\theta_2$

$$1.00 \sin 30.0° = n \sin 19.24°$$

$$n = \boxed{1.52}$$

(c) $$f = \frac{c}{\lambda} = \frac{3.00 \times 10^8\ \text{m/s}}{6.328 \times 10^{-7}\ \text{m}} = \boxed{4.74 \times 10^{14}\ \text{Hz}}$$ in air and in syrup.

(d) $$v = \frac{c}{n} = \frac{3.00 \times 10^8\ \text{m/s}}{1.52} = 1.98 \times 10^8\ \text{m/s} = \boxed{198\ \text{Mm/s}}$$

(b) $$\lambda = \frac{v}{f} = \frac{1.98 \times 10^8\ \text{m/s}}{4.74 \times 10^{14}\ \text{s}^{-1}} = \boxed{417\ \text{nm}}$$

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***P35.17** (a) The angle of incidence at the first surface is $\theta_{1i} = \boxed{30.0°}$, and the angle of refraction is

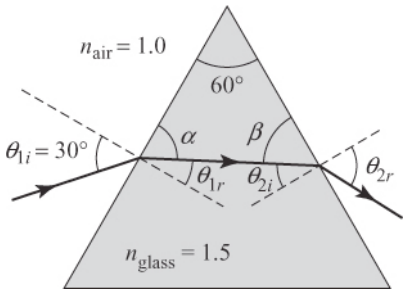
$$\theta_{1r} = \sin^{-1}\left(\frac{n_{\text{air}} \sin\theta_{1i}}{n_{\text{glass}}}\right) = \sin^{-1}\left(\frac{1.0 \sin 30°}{1.5}\right) = \boxed{19°}$$

Also, $\alpha = 90° - \theta_{1r} = 71°$ and $\beta = 180° - 60° - \alpha = 49°$.

Therefore, the angle of incidence at the second surface is $\theta_{2i} = 90° - \beta = \boxed{41°}$. The angle of refraction at this surface is

$$\theta_{2r} = \sin^{-1}\left(\frac{n_{\text{glass}} \sin\theta_{2i}}{n_{\text{air}}}\right) = \sin^{-1}\left(\frac{1.5 \sin 41°}{1.0}\right) = \boxed{77°}$$

ANS. FIG. P35.17 traces the path of the ray of light.

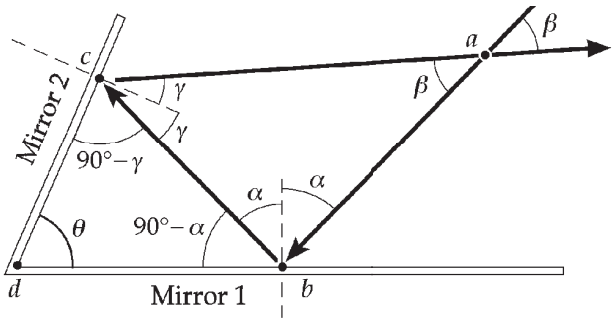


ANS. FIG. P35.17

(b) The angle of reflection at each surface equals the angle of incidence at that surface. Thus,

$$(\theta_1)_{\text{reflection}} = \theta_{1i} = \boxed{30°}, \text{ and } (\theta_1)_{\text{reflection}} = \theta_{2i} = \boxed{41°}$$

***P35.18** ANS. FIG. P35.18 shows the path of the light ray. α and γ are angles of incidence at mirrors 1 and 2.



ANS. FIG. P35.18

For triangle *abca*,

$$2\alpha + 2\gamma + \beta = 180°$$

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or $\beta = 180° - 2(\alpha + \gamma).$ **[1]**

Now for triangle *bcdb*,

$$(90.0° - \alpha) + (90.0° - \gamma) + \theta = 180°$$

or $\theta = \alpha + \gamma.$ **[2]**

Substituting equation [2] into equation [1] gives $\boxed{\beta = 180° - 2\theta}$.

Note: From equation [2], $\gamma = \theta - \alpha$. Thus, the ray will follow a path like that shown only if $\alpha < \theta$. For $\alpha > \theta$, γ is negative and multiple reflections from each mirror will occur before the incident and reflected rays intersect.

***P35.19** Consider glass with an index of refraction of 1.50, which is 3.00 mm thick. The speed of light in the glass is

$$\frac{3.00 \times 10^8 \text{ m/s}}{1.50} = 2.00 \times 10^8 \text{ m/s}$$

The extra travel time is

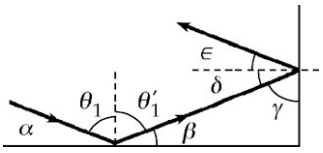
$$\frac{3.00 \times 10^{-3} \text{ m}}{2.00 \times 10^8 \text{ m/s}} - \frac{3.00 \times 10^{-3} \text{ m}}{3.00 \times 10^8 \text{ m/s}} \boxed{\sim 10^{-11} \text{ s}}$$

For light of wavelength 600 nm in vacuum and wavelength $\frac{600 \text{ nm}}{1.50} = 400 \text{ nm}$ in glass, the extra optical path, in wavelengths, is

$$\frac{3 \times 10^{-3} \text{ m}}{4 \times 10^{-7} \text{ m}} - \frac{3 \times 10^{-3} \text{ m}}{6 \times 10^{-7} \text{ m}} \boxed{\sim 10^3 \text{ wavelengths}}$$

P35.20 (a) **Method One:**

The incident ray makes angle $\alpha = 90° - \theta_1$ with the first mirror. In ANS. FIG. P35.20, the law of reflection implies that $\theta_1 = \theta_1'$



ANS. FIG. P35.20

Then,

$$\beta = 90° - \theta_1' = 90 - \theta_1 = \alpha.$$

In the triangle made by the mirrors and the ray passing between them,

$$\beta + 90° + \gamma = 180°$$

$$\gamma = 90° - \beta$$

Further, $\delta = 90° - \gamma = \beta = \alpha$

and $\epsilon = \delta = \alpha.$

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Thus the final ray makes the same angle with the first mirror as did the incident ray. Its direction is opposite to the incident ray.

Method Two:

The vector velocity of the incident light has a component v_y perpendicular to the first mirror and a component v_x perpendicular to the second. The v_y component is reversed upon the first reflection, which leaves v_x unchanged. The second reflection reverses v_x and leaves v_y unchanged. The doubly reflected ray then has velocity opposite to the incident ray.

(b) The incident ray has velocity $v_x\hat{\mathbf{i}} + v_y\hat{\mathbf{j}} + v_z\hat{\mathbf{k}}$. If all of these components are non-zero, the light will reflect from each mirror because each component carries the light into the mirror that is perpendicular to that component: for example, the x component of velocity carries the light into the mirror in the yz plane. Each reflection reverses one component and leaves the other two unchanged. After all the reflections, the light has velocity $-v_x\hat{\mathbf{i}} - v_y\hat{\mathbf{j}} - v_z\hat{\mathbf{k}}$, opposite to the incident ray.

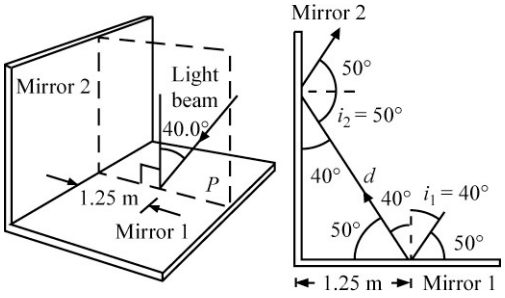
P35.21 (a) From geometry,

$$1.25 \text{ m} = d \sin 40.0°$$

so $d = \boxed{1.94 \text{ m}}$.

(b) $\boxed{50.0° \text{ above the horizontal}}$

or parallel to the incident ray.



ANS. FIG. P35.21

P35.22 (a) At entry, $n_1 \sin\theta_1 = n_2 \sin\theta_2$,

or $1.00 \sin 30.0° = 1.50 \sin\theta_2$,

which gives $\theta_2 = 19.5°$.

The distance h the light travels in the medium is given by

$$\cos\theta_2 = \frac{2.00 \text{ cm}}{h}$$

or $$h = \frac{2.00 \text{ cm}}{\cos 19.5°} = 2.12 \text{ cm}.$$

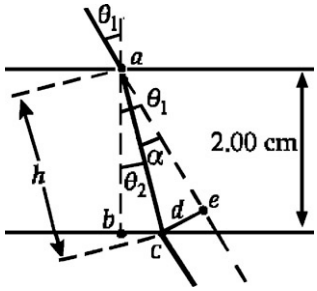
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The angle of deviation upon entry is

$$\alpha = \theta_1 - \theta_2 = 30.0° - 19.5° = 10.5°$$

The offset distance comes from $\sin\alpha = \frac{d}{h}$:

$$d = (2.12\text{ cm})\sin 10.5° = \boxed{0.387\text{ cm}}$$



ANS. FIG. P35.22

(b) The speed of light in the material is

$$v = \frac{c}{n} = \frac{3.00\times10^8\text{ m/s}}{1.50} = 2.00\times10^8\text{ m/s}$$

The distance h traveled by the light is $h = 2.12$ cm. The time interval is

$$\Delta t = \frac{h}{v} = \frac{2.12\times10^{-2}\text{ m}}{2.00\times10^8\text{ m/s}} = 1.06\times10^{-10}\text{ s} = \boxed{106\text{ ps}}$$

P35.23 From Table 35.1, the index of refraction of ice is 1.309. The pulses are in step with each other until one enters the ice, then that pulse slows down. The difference in the times of arrival of the pulses is

$$\Delta t = \frac{L}{v_{\text{ice}}} - \frac{L}{v_{\text{air}}} = \frac{L}{c/n_{\text{ice}}} - \frac{L}{c/n_{\text{air}}} = (n_{\text{ice}} - n_{\text{air}})\frac{L}{c}$$

$$\Delta t = (1.309 - 1.000)\frac{6.20\text{ m}}{3.00\times10^8\text{ m/s}} = 6.39\times10^{-9}\text{ s} = \boxed{6.39\text{ ns}}$$

P35.24 Refraction proceeds according to

$$(1.00)\sin\theta_1 = (1.66)\sin\theta_2 \qquad \textbf{[1]}$$

(a) For the normal component of velocity to be constant,

$$v_1\cos\theta_1 = v_2\cos\theta_2 \quad \text{or} \quad (c)\cos\theta_1 = \left(\frac{c}{1.66}\right)\cos\theta_2 \qquad \textbf{[2]}$$

We multiply equations [1] and [2], obtaining:

$$\sin\theta_1\cos\theta_1 = \sin\theta_2\cos\theta_2 \quad \text{or} \quad \sin 2\theta_1 = \sin 2\theta_2$$

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We do not consider the case $\theta_1 = 0$. The physical solution is

$$2\theta_1 = 180° - 2\theta_2 \quad \text{or} \quad \theta_2 = 90.0° - \theta_1$$

Then equation [1] becomes:

$$\sin\theta_1 = 1.66\cos\theta_1$$
$$\tan\theta_1 = 1.66$$
$$\theta_1 = 58.9°$$

Yes, if the angle of incidence is 58.9°.

(b) No. Both the reduction in speed and the bending toward the normal reduce the component of velocity parallel to the interface. This component cannot remain constant for a nonzero angle of incidence.

P35.25 (a) As measured from the diagram, the incidence angle is 60°, and the refraction angle is 35°. From Snell's law, $\frac{\sin\theta_2}{\sin\theta_1} = \frac{n_1}{n_2} = \frac{v_2}{v_1}$, then $\frac{\sin 35°}{\sin 60°} = \frac{v_2}{c}$ and the speed of light in the block is $\boxed{2.0\times10^8 \text{ m/s}}$.

(b) The frequency of the light does not change upon refraction. Knowing the wavelength in a vacuum, we can use the speed of light in a vacuum to determine the frequency: $c = f\lambda$, thus $3.00\times10^8 = f\left(632.8\times10^{-9}\right)$, so the frequency is $\boxed{4.74\times10^{14} \text{ Hz}}$.

(c) To find the wavelength of light in the block, we use the same wave speed relation, $v = f\lambda$, so $2.0\times10^8 = \left(4.74\times10^{14}\right)\lambda$, so $\lambda_{\text{glass}} = 4.20\times10^{-7} = \boxed{420 \text{ nm}}$.

P35.26 From Snell's law, the angle of refraction θ inside the liver is

$$\sin\theta = \left(\frac{n_{\text{medium}}}{n_{\text{liver}}}\right)\sin 50.0°$$

But $$\frac{n_{\text{medium}}}{n_{\text{liver}}} = \frac{c/v_{\text{medium}}}{c/v_{\text{liver}}} = \frac{v_{\text{liver}}}{v_{\text{medium}}} = 0.900,$$

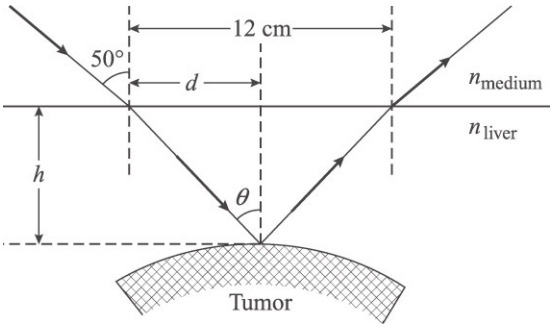
so $$\theta = \sin^{-1}\left[(0.900)\sin 50.0°\right] = 43.6°.$$

From the law of reflection,

$$d = \frac{12.0 \text{ cm}}{2} = 6.00 \text{ cm}$$

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and $$h = \frac{d}{\tan\theta} = \frac{6.00\text{ cm}}{\tan 43.6°} = \boxed{6.30\text{ cm}}.$$



ANS. FIG. P35.26

P35.27 The refracted sunlight does not illuminate any part of the bottom when it strikes its far inside edge:

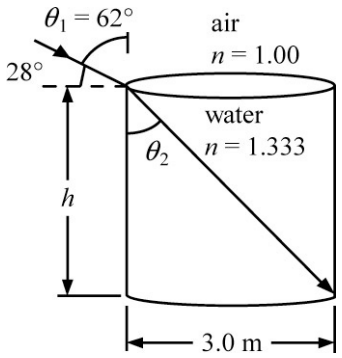
$$\sin\theta_1 = n_w \sin\theta_2$$

$$\sin\theta_2 = \frac{1}{1.333}\sin\theta_1$$

$$= \frac{1}{1.333}\sin(90.0° - 28.0°) = 0.662$$

$$\theta_2 = \sin^{-1}(0.662) = 41.5°$$

$$h = \frac{d}{\tan\theta_2} = \frac{3.00\text{ m}}{\tan 41.5°} = \boxed{3.39\text{ m}}$$



ANS. FIG. P35.27

P35.28 Note for use in every part (refer to ANS. FIG. P35.28): from apex angle Φ,

$$\Phi + (90.0° - \theta_2) + (90.0° - \theta_3) = 180°$$

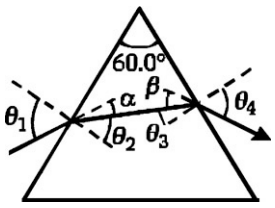
so $$\theta_3 = \Phi - \theta_2$$

At the first surface the deviation is

$$\alpha = \theta_1 - \theta_2$$

At exit, the deviation is

$$\beta = \theta_4 - \theta_3$$



ANS. FIG. P35.28

The total deviation is therefore

$$\delta = \alpha + \beta = \theta_1 + \theta_4 - \theta_2 - \theta_3 = \theta_1 + \theta_4 - \Phi$$

(a) At entry,

$$n_1 \sin\theta_1 = n_2 \sin\theta_2 \quad \text{or} \quad \theta_2 = \sin^{-1}\left(\frac{\sin 48.6°}{1.50}\right) = 30.0°$$

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Thus, $\theta_3 = 60.0° - 30.0° = 30.0°$

At exit,

$$1.50 \sin 30.0° = 1.00 \sin \theta_4$$

or $\theta_4 = \sin^{-1}\left[1.50 \sin(30.0°)\right] = 48.6°$

so the path through the prism is symmetric when $\theta_1 = 48.6°$.

(b) $\delta = 48.6° + 48.6° - 60.0° = \boxed{37.2°}$

(c) At entry,

$$\sin \theta_2 = \frac{\sin 45.6°}{1.50} \Rightarrow \theta_2 = 28.4°$$

$$\theta_3 = 60.0° - 28.4° = 31.6°$$

At exit,

$$\sin \theta_4 = 1.50 \sin(31.6°) \Rightarrow \theta_4 = 51.7°$$

$$\delta = 45.6° + 51.7° - 60.0° = \boxed{37.3°}$$

(d) At entry,

$$\sin \theta_2 = \frac{\sin 51.6°}{1.50} \Rightarrow \theta_2 = 31.5°$$

$$\theta_3 = 60.0° - 31.5° = 28.5°$$

At exit,

$$\sin \theta_4 = 1.50 \sin(28.5°) \Rightarrow \theta_4 = 45.7°$$

$$\delta = 51.6° + 45.7° - 60.0° = \boxed{37.3°}$$

P35.29 The index of refraction at 700 nm is $n(700 \text{ nm}) = 1.458$.

(a) $(1.00)\sin 75.0° = 1.458 \sin \theta_2$; $\theta_2 = \boxed{41.5°}$

(b) Refer to ANS. FIG. P35.29. Let

$$\theta_3 + \beta = 90.0° \text{ and } \theta_2 + \alpha = 90.0°$$

then,

$$\alpha + \beta + 60.0° = 180°$$

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So

$$\alpha + \beta + 60.0° = 180°$$

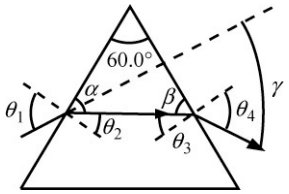
$$(90.0° - \theta_2) + (90.0° - \theta_3) + 60.0° = 180°$$

$$60.0° - \theta_2 - \theta_3 = 0 \Rightarrow 60.0° - 41.5° = \theta_3 = \boxed{18.5°}$$

(c) $1.458 \sin 18.5° = 1.00 \sin \theta_4 \quad \rightarrow \quad \theta_4 = \boxed{27.6°}$

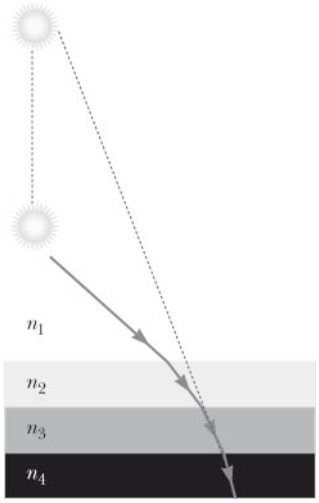
(d) $\gamma = (\theta_1 - \theta_2) + (\theta_4 - \theta_3)$

$$\gamma = (75.0° - 41.5°) + (27.6° - 18.5°) = \boxed{42.6°}$$



ANS. FIG. P35.29

P35.30 The index of refraction of the atmosphere decreases with increasing altitude because of the decrease in density of the atmosphere with increasing altitude. As indicated in the ray diagram, the sun located at S below the horizon appears to be located at S′.



ANS. FIG. P35.30

P35.31 For sheets 1 and 2 as described,

$$n_1 \sin 26.5° = n_2 \sin 31.7°$$

$$0.849 n_1 = n_2$$

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For the trial with sheets 3 and 2,

$$n_3 \sin 26.5° = n_2 \sin 36.7°$$
$$0.747 n_3 = n_2$$

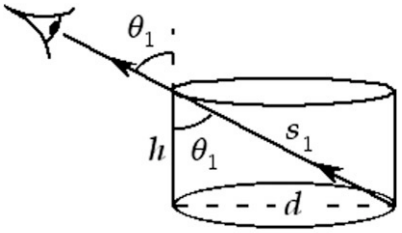
Equate the two expressions for n_2:

$$0.747 n_3 = 0.849 n_1$$
$$n_3 = 1.14 n_1$$

For the third trial,

$$n_1 \sin 26.5° = n_3 \sin \theta_3 = 1.14 n_1 \sin \theta_3$$
$$\theta_3 = \boxed{23.1°}$$

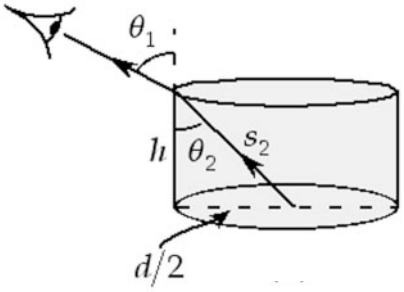
P35.32 (a) Before the container is filled, the ray's path is as shown in ANS. FIG. P35.32(a). From this figure, observe that



ANS. FIG. P35.32(a)

$$\sin \theta_1 = \frac{d}{s_1} = \frac{d}{\sqrt{h^2 + d^2}}$$
$$= \frac{1}{\sqrt{(h/d)^2 + 1}}$$

After the container is filled, the ray's path is shown in ANS. FIG. P35.32(b). From this figure, we find that



ANS. FIG. P35.32(b)

$$\sin \theta_2 = \frac{d/2}{s_2} = \frac{d/2}{\sqrt{h^2 + (d/2)^2}}$$
$$= \frac{1}{\sqrt{4(h/d)^2 + 1}}$$

From Snell's law, we have

$$1.00 \sin \theta_1 = n \sin \theta_2$$
$$\frac{1.00}{\sqrt{(h/d)^2 + 1}} = \frac{n}{\sqrt{4(h/d)^2 + 1}}$$
$$4(h/d)^2 + 1 = n^2 (h/d)^2 + n^2$$
$$(h/d)^2 (4 - n^2) = n^2 - 1 \rightarrow \boxed{\frac{h}{d} = \sqrt{\frac{n^2 - 1}{4 - n^2}}}$$

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(b) For water, $n = 1.333$.

$$\frac{h}{d} = \sqrt{\frac{n^2 - 1}{4 - n^2}}$$

$$\frac{h}{8.00\text{ cm}} = \sqrt{\frac{(1.333)^2 - 1}{4 - (1.333)^2}} = \boxed{4.73\text{ cm}}$$

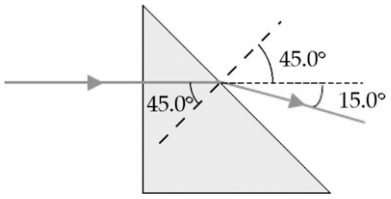
(c) $\boxed{\text{For } n = 1,\ h = 0.\ \text{For } n = 2,\ h = \infty.\ \text{For } n > 2,\ h \text{ has no real solution.}}$

P35.33 Since the light ray strikes the first surface at normal incidence, it passes into the prism without deviation. Thus, the angle of incidence at the second surface (hypotenuse of the triangular prism) is $\theta_1 = 45.0°$ as shown in the sketch at the right. The angle of refraction is

$$\theta_2 = 45.0° + 15.0° = 60.0°$$

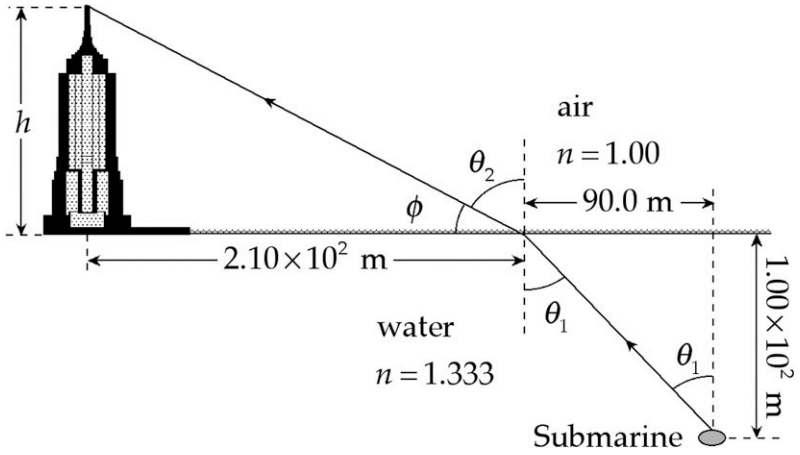
and Snell's law gives the index of refraction of the prism material as

$$n_1 = \frac{n_2 \sin\theta_2}{\sin\theta_1} = \frac{(1.00)\sin(60.0°)}{\sin(45.0°)} = \boxed{1.22}$$



ANS. FIG. P35.33

P35.34 (a) A sketch illustrating the situation and the two triangles needed in the solution is given in ANS. FIG. P35.34.



ANS. FIG. P35.34

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(b) From the triangle under water, the angle of incidence θ_1 at the water surface is

$$\tan\theta_1 = \frac{90.0\text{ m}}{100\text{ m}} \quad \rightarrow \quad \theta_1 = \boxed{42.0^\circ}$$

(c) Snell's law gives the angle of refraction as

$$\theta_2 = \sin^{-1}\left(\frac{n_{\text{water}}\sin\theta_1}{n_{\text{air}}}\right) = \sin^{-1}\left(\frac{(1.333)\sin 42.0^\circ}{1.00}\right) = \boxed{63.1^\circ}$$

(d) The refracted beam makes angle $\phi = 90.0^\circ - \theta_2 = \boxed{26.9^\circ}$ with the horizontal.

(e) In the triangle above the water,

$$h = (210\text{ m})\tan\phi = (210\text{ m})\tan 26.9^\circ = \boxed{107\text{ m}}$$

P35.35 The reflected ray and refracted ray are perpendicular to each other, and the angle of reflection θ_1 and the angle of refraction θ_2 are related by

$$\theta_1 + 90.0^\circ + \theta_2 = 180.0^\circ \quad \rightarrow \quad \theta_2 = 90.0^\circ - \theta_1$$

Then, from Snell's law,

$$\sin\theta_1 = \frac{n_g\sin\theta_2}{n_{\text{air}}}$$

$$= n_g\sin\left(90^\circ - \theta_1\right) = n_g\cos\theta_1$$

Thus, $\dfrac{\sin\theta_1}{\cos\theta_1} = \tan\theta_1 = n_g$ or $\boxed{\theta_1 = \tan^{-1}\left(n_g\right)}$

Section 35.6 Huygens's Principle

Section 35.7 Dispersion

P35.36 Using Snell's law gives

(a) $$\theta_{\text{red}} = \sin^{-1}\left(\frac{n_{\text{air}}\sin\theta_i}{n_{\text{red}}}\right) = \sin^{-1}\left(\frac{(1.000)\sin 83.0^\circ}{1.331}\right) = \boxed{48.2^\circ}$$

(b) $$\theta_{\text{blue}} = \sin^{-1}\left(\frac{n_{\text{air}}\sin\theta_i}{n_{\text{blue}}}\right) = \sin^{-1}\left(\frac{(1.000)\sin 83.0^\circ}{1.340}\right) = \boxed{47.8^\circ}$$

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P35.37 Using Snell's law gives

$$\theta_{\text{red}} = \sin^{-1}\left(\frac{n_{\text{air}}\sin\theta_i}{n_{\text{red}}}\right) = \sin^{-1}\left(\frac{(1.000)\sin 50.00°}{1.455}\right)$$

and $$\theta_{\text{violet}} = \sin^{-1}\left(\frac{n_{\text{air}}\sin\theta_i}{n_{\text{violet}}}\right) = \sin^{-1}\left(\frac{(1.000)\sin 50.00°}{1.468}\right)$$

Thus, the dispersion is $\theta_{\text{red}} - \theta_{\text{violet}} = \boxed{0.314°}$

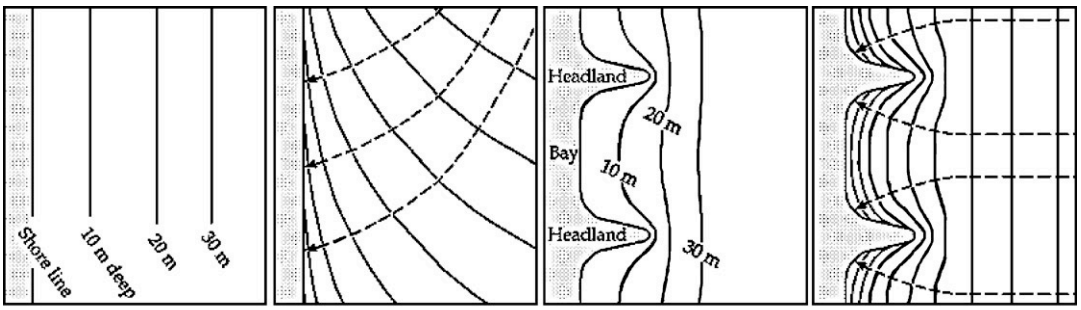
P35.38 Recall that if a wave slows down as it passes from one medium into another, its rays tend to bend toward the normal, unless it has normal incidence. Example: the case when light passes from air into water.

(a) For the diagrams of contour lines and wave fronts and rays, see ANS. FIG. P35.38(a) below.

(b) As the waves move to shallower water, the wave fronts slow down, and those closer to shore slow down more. The rays tend to bend toward the normal of the contour lines; or equivalently, the wave fronts bend to become more nearly parallel to the contour lines. See ANS. FIG. P35.38(b) below.

(c) For the diagrams of contour lines and wave fronts and rays, see ANS. FIG. P35.38(c) below.

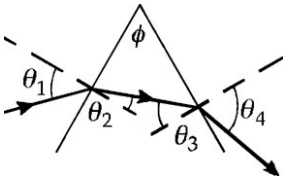
(d) We suppose that the headlands are steep underwater, as they are above water. The rays are everywhere perpendicular to the wave fronts of the incoming refracting waves. As shown, because the rays tend to bend toward the normal of the contour lines, the rays bend toward the headlands and deliver more energy per length at the headlands. See ANS. FIG. P35.38(d) below.



ANS. FIG. P35.38

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P35.39 For the incoming ray, $\sin\theta_2 = \dfrac{\sin\theta_1}{n}$.



ANS. FIG. P35.39

Using ANS. FIG. P35.39,

$$(\theta_2)_{\text{violet}} = \sin^{-1}\left(\frac{\sin 50.0^\circ}{1.66}\right) = 27.48^\circ$$

$$(\theta_2)_{\text{red}} = \sin^{-1}\left(\frac{\sin 50.0^\circ}{1.62}\right) = 28.22^\circ$$

For the outgoing ray,

$$(90.0^\circ - \theta_2) + (90.0^\circ - \theta_3) + 60.0^\circ = 180.0^\circ$$

$$\theta_3 = 60.0^\circ - \theta_2$$

and

$$\sin\theta_4 = n\sin\theta_3: \qquad (\theta_4)_{\text{violet}} = \sin^{-1}[1.66\sin 32.52^\circ] = 63.17^\circ$$

$$(\theta_4)_{\text{red}} = \sin^{-1}[1.62\sin 31.78^\circ] = 58.56^\circ$$

The angular dispersion is the difference

$$\Delta\theta_4 = (\theta_4)_{\text{violet}} - (\theta_4)_{\text{red}} = 63.17^\circ - 58.56^\circ = \boxed{4.61^\circ}$$

P35.40 For the incoming ray, $\sin\theta_2 = \dfrac{\sin\theta_1}{n}$. Using ANS. FIG. P35.40,

$$(\theta_2)_{\text{violet}} = \sin^{-1}\left(\frac{\sin\theta}{n_V}\right)$$

$$(\theta_2)_{\text{red}} = \sin^{-1}\left(\frac{\sin\theta}{n_R}\right)$$

For the outgoing ray,

$$(90.0^\circ - \theta_2) + (90.0^\circ - \theta_3) + \Phi = 180.0^\circ$$

$$\theta_3 = \Phi - \theta_2$$

and

$$\sin\theta_4 = n\sin\theta_3: \quad (\theta_4)_{\text{violet}} = \sin^{-1}\left\{n_V \sin\left[\Phi - \sin^{-1}\left(\frac{\sin\theta}{n_V}\right)\right]\right\}$$

$$(\theta_4)_{\text{red}} = \sin^{-1}\left\{n_R \sin\left[\Phi - \sin^{-1}\left(\frac{\sin\theta}{n_R}\right)\right]\right\}$$

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The angular dispersion is the difference

$$\Delta\theta_4 = (\theta_4)_{\text{violet}} - (\theta_4)_{\text{red}}$$

$$= \boxed{\begin{aligned}&\sin^{-1}\left\{n_V \sin\left[\Phi - \sin^{-1}\left(\frac{\sin\theta}{n_V}\right)\right]\right\} \\ &\qquad - \sin^{-1}\left\{n_R \sin\left[\Phi - \sin^{-1}\left(\frac{\sin\theta}{n_R}\right)\right]\right\}\end{aligned}}$$

Section 35.8 Total Internal Reflection

P35.41 From Equation 35.10,

$$\sin\theta_c = \frac{n_2}{n_1} = \frac{1.33}{1.50} \rightarrow \theta_c = \boxed{62.5^\circ}$$

P35.42 From Equation 35.10, $\sin\theta_c = \dfrac{n_2}{n_1}$, where $n_2 = 1.000\ 293$. Values for n_1 come from Table 35.1,

(a) $\theta_c = \sin^{-1}\left(\dfrac{1.000\ 293}{2.20}\right) = \boxed{27.0^\circ}$

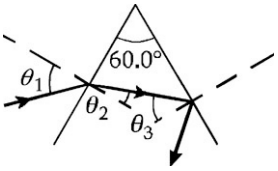
(b) $\theta_c = \sin^{-1}\left(\dfrac{1.000\ 293}{1.66}\right) = \boxed{37.1^\circ}$

(c) $\theta_c = \sin^{-1}\left(\dfrac{1.000\ 293}{1.309}\right) = \boxed{49.8^\circ}$

P35.43 The prism is in air, so at the first refraction,

$$1.00\sin\theta_1 = n\sin\theta_2$$

The angle of incidence θ_3 must be less than the critical angle at the second surface to emerge from the other side.



ANS. FIG. P35.43

$$\theta_3 < \theta_c$$

$$\theta_3 < \sin^{-1}\theta_c = \sin^{-1}\frac{n_2}{n_1} = \sin^{-1}\left(\frac{1.00}{1.50}\right)$$

$$\theta_3 < 41.8^\circ$$

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The angles θ_2 and θ_3 are related by

$$(90.0° - \theta_2) + (90.0° - \theta_3) + 60.0° = 180.0°$$

$$\theta_2 = 60.0° - \theta_3$$

Thus, to avoid total internal reflection at the second surface (i.e., have $\theta_3 < 41.8°$), it is necessary that $\theta_2 > 18.2°$.

Since $\sin\theta_1 = n\sin\theta_2$, this becomes

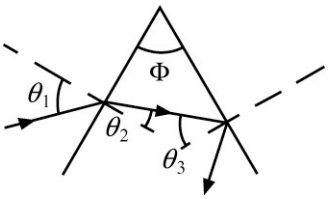
$$\sin\theta_1 > 1.50\sin 18.2° = 0.468$$

or $$\theta_1 > \boxed{27.9°}.$$

P35.44 The prism is in air, so at the first refraction,

$$1.00\sin\theta_1 = n\sin\theta_2$$

The angle of incidence θ_3 must be less than the critical angle at the second surface to emerge from the other side.



ANS. FIG. P35.44

$$\theta_3 < \theta_c$$

$$\theta_3 < \sin^{-1}\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right) = \sin^{-1}\left(\frac{1.00}{n}\right)$$

The angles θ_2 and θ_3 are related:

$$(90.0° - \theta_2) + (90.0° - \theta_3) + \Phi = 180°$$

which gives $\theta_2 = \Phi - \theta_3$.

Thus, to have $\theta_3 < \sin^{-1}\left(\frac{1.00}{n}\right)$ and avoid total internal reflection at the second surface, it is necessary that $\theta_2 > \Phi - \sin^{-1}\left(\frac{1.00}{n}\right)$.

Since $\sin\theta_1 = n\sin\theta_2$, this requirement becomes

$$\sin\theta_1 > n\sin\left[\Phi - \sin^{-1}\left(\frac{1.00}{n}\right)\right]$$

or $$\theta_1 > \boxed{\sin^{-1}\left(n\sin\left[\Phi - \sin^{-1}\left(\frac{1.00}{n}\right)\right]\right)}.$$

Through the application of trigonometric identities,

$$\theta_1 > \boxed{\sin^{-1}\left(\sqrt{n^2 - 1}\sin\Phi - \cos\Phi\right)}$$

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P35.45 At the upper surface,

$$\sin\theta_c = \frac{n_{\text{air}}}{n_{\text{pipe}}} = \frac{1.00}{1.36} = 0.735 \quad \rightarrow \quad \theta_c = 47.3°$$

Geometry shows that the angle of refraction at the end is

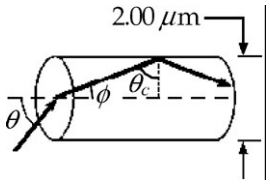
$$\phi = 90.0° - \theta_c = 90.0° - 47.3° = 42.7°$$

Then, by Snell's law at the end,

$$1.00\sin\theta = 1.36\sin 42.7°$$

gives $\boxed{\theta = 67.2°}$.

The 2-μm diameter is unnecessary information.



ANS. FIG. P35.45

P35.46 (a) Using the index of refraction values listed in Table 35.1, we find

$$\sin\theta_c = \frac{n_2}{n_1} = \frac{1.000}{2.419} \rightarrow \theta_c = \boxed{24.42°}$$

(b) Because the angle of incidence (35.0°) is greater than the critical angle, the light is totally reflected at P.

(c) $\sin\theta_c = \dfrac{n_2}{n_1} = \dfrac{1.333}{2.419} \rightarrow \theta_c = \boxed{33.44°}$

(d) The angle of incidence is 35.0°. Yes. In this case, the angle of incidence is just larger than the critical angle, so the light ray again undergoes total internal reflection at P.

(e) The angle of incidence must be reduced below the critical angle for light to exit the diamond, so the diamond should be rotated clockwise.

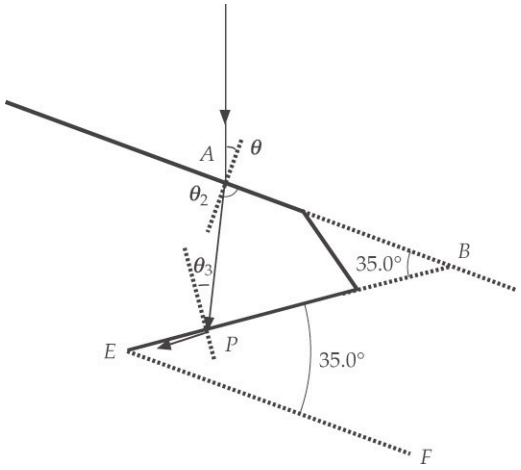
(f) Rotating the diamond by angle θ clockwise changes the angle of incidence θ_1 at point A from 0.00° to θ, causing the angle of refraction θ_2 inside the diamond to change from 0.00°:

$$n_1\sin\theta_1 = n_2\sin\theta_2$$

$$1.333\sin\theta_1 = 2.419\sin\theta_2$$

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Refer to ANS. FIG. P35.46. What is the angle of incidence at *P*? Extending a line from points *A* and *P* parallel to the surfaces of the diamond until they meet at point *B*, we form a triangle *ABP*.



ANS. FIG. P35.46

The angle at vertex *B* is 35.0° because the extended line *AB* is parallel to the line *EF* extended from the base of the diamond. From the sum of the interior angles of *ABP*, we find the incident angle θ_3 at point *P*:

$$(90.0° - \theta_2) + (90.0° - \theta_3) + 35.0° = 180$$

$$\theta_3 = 35.0° - \theta_2$$

At *P*, we require that the angle of incidence θ_3 results in an angle of refraction of 90.0°:

$$2.419 \sin\theta_3 = 1.333 \sin 90.0°$$

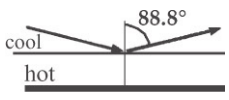
$$2.419 \sin(35.0° - \theta_2) = 1.333$$

$$35.0° - \theta_2 = \sin^{-1}\frac{1.333}{2.419}$$

solving gives $\theta_2 = 1.561°$. Then, from above,

$$1.333 \sin\theta_1 = 2.419 \sin\theta_2 \;\rightarrow\; \theta = \boxed{2.83°}$$

P35.47 The line of sight is 1.20° below the horizontal, so the angle of reflection of the light reaching the truck driver's eyes is 90.0° – 1.20° = 88.8°.



ANS. FIG. P35.47

$$\sin\theta_c = \frac{n_2}{n_1}$$

$$n_2 = n_1 \sin 88.8° = (1.000\,293)\sin 88.8° = \boxed{1.000\,07}$$

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Note: Mirages are caused by a continuous variation in index of refraction of the air rather than by total internal reflection. In this problem, the intent is to recognize that the result of the variation in index of refraction is equivalent to the result of a total internal reflection occurring at a single layer of hot air just above the surface of the roadway. This problem MODELS the phenomenon as a total internal reflection.

P35.48 (a) $\dfrac{\sin\theta_2}{\sin\theta_1}=\dfrac{v_2}{v_1}$ and $\theta_2 = 90.0°$ at the critical angle.

$$\frac{\sin 90.0°}{\sin\theta_c}=\frac{1\,850\text{ m/s}}{343\text{ m/s}} \quad \text{so} \quad \theta_c = \sin^{-1}(0.185) = \boxed{10.7°}.$$

(b) Sound can be totally reflected if it is traveling in the medium where it travels slower: $\boxed{\text{air}}$.

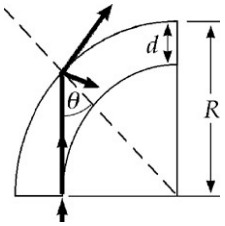
(c) $\boxed{\text{Sound in air falling on the wall from most directions is 100\% reflected,}}$

so the wall is a good mirror.

P35.49 (a) If any ray escapes it will be a ray along the inner edge, because it has the smallest angle of incidence. Its angle of incidence is described by $\sin\theta = \dfrac{R-d}{R}$ and by $n\sin\theta > 1\sin 90°$. Then

$$\frac{n(R-d)}{R} > 1 \quad \rightarrow \quad nR - nd > R$$

$$\rightarrow \quad nR - R > nd \quad \rightarrow \quad R > \boxed{\frac{nd}{n-1}}$$



ANS. FIG. P35.49

(b) As $d \rightarrow 0$, $\boxed{R_{\min} \rightarrow 0}$. $\boxed{\text{Yes: for very small } d\text{, the light strikes the interface at very large angles of incidence.}}$

(c) As n increases, $\boxed{R_{\min}}$ decreases. Yes: as n increases, the critical angle becomes smaller.

(d) As n decreases toward 1, $R_{\min}$ increases. $\boxed{R_{\min} \rightarrow \infty}$. Yes: as $n \rightarrow 1$, the critical angle becomes close to 90° and any bend will allow the light to escape.

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(e) $$R_{\min} = \frac{1.40\left(100 \times 10^{-6}\ \text{m}\right)}{0.40} = 350 \times 10^{-6}\ \text{m} = \boxed{350\ \mu\text{m}}$$

P35.50 (a) In the gasoline gauge, skylight from above travels down the plastic. The rays close to the vertical are totally reflected from the sides of the slab and from both facets at the lower end of the plastic, where it is not immersed in gasoline. This light returns up inside the plastic and makes it look bright. Where the plastic is immersed in gasoline, with index of refraction of about 1.50, total internal reflection should not happen. The light passes out of the lower end of the plastic with little reflected, making this part of the gauge look dark.

(b) To ensure total internal reflection at the plastic-air interface, the critical angle must be less than the angle of incidence, about 45.0°. This places a lower limit on the index of refraction of the plastic:

$$\theta_c \le 45.0°$$
$$\sin\theta_c \le \sin 45.0°$$
$$\frac{1}{n} \le \sin 45.0° \rightarrow \boxed{n \ge 1.41}$$

To prevent total internal reflection at the plastic-gasoline interface, the critical angle must be greater than the angle of incidence. This places an upper limit on the index of refraction of the plastic:

$$\theta_c \ge 45.0°$$
$$\sin\theta_c \ge \sin 45.0°$$
$$\frac{1.50}{n} \ge \sin 45.0° \rightarrow \boxed{n \le 2.12}$$

Additional Problems

***P35.51** Using Snell's law, the index of refraction of the liquid is found to be

$$n_{\text{liquid}} = \frac{n_{\text{air}} \sin\theta_i}{\sin\theta_r}$$

Thus, the critical angle for light going from this liquid into air is

$$\theta_c = \sin^{-1}\left(\frac{n_{\text{air}}}{n_{\text{liquid}}}\right) = \sin^{-1}\left(\frac{n_{\text{air}}}{n_{\text{air}} \sin\theta_i / \sin\theta_r}\right)$$
$$= \sin^{-1}\left(\frac{\sin\theta_r}{\sin\theta_i}\right) = \sin^{-1}\left(\frac{\sin 22.0°}{\sin 30.0°}\right) = \boxed{48.5°}$$

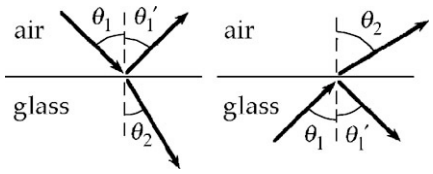
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P35.52 (a) $\theta_1' = \theta_1 = \boxed{30.0^\circ}$

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

$$1.00 \sin 30.0^\circ = 1.55 \sin\theta_2$$

$$\theta_2 = \boxed{18.8^\circ}$$



ANS. FIG. P35.52

(b) $\theta_1' = \theta_1 = \boxed{30.0^\circ}$

$$\theta_2 = \sin^{-1}\left(\frac{n_1 \sin\theta_1}{n_2}\right) = \sin^{-1}\left(\frac{1.55 \sin 30.0^\circ}{1}\right) = \boxed{50.8^\circ}$$

(c), (d) The other entries are computed similarly, and are shown in Table P35.52 below.

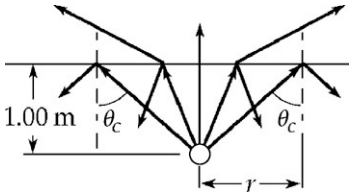
(c) air into glass, angles in degrees			(d) glass into air, angles in degrees		
incidence	reflection	refraction	incidence	reflection	refraction
0	0	0	0	0	0
10.0	10.0	6.43	10.0	10.0	15.6
20.0	20.0	12.7	20.0	20.0	32.0
30.0	30.0	18.8	30.0	30.0	50.8
40.0	40.0	24.5	40.0	40.0	85.1
50.0	50.0	29.6	50.0	50.0	none*
60.0	60.0	34.0	60.0	60.0	none*
70.0	70.0	37.3	70.0	70.0	none*
80.0	80.0	39.4	80.0	80.0	none*
90.0	90.0	40.2	90.0	90.0	none*

*total internal reflection

TABLE P35.52

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P35.53 The critical angle is found by imagining the refracted ray just grazing the surface (θ_2 = 90°). The index of refraction of water is n_1 = 1.333, and $n_2 = 1.00$ for air, so $n_1 \sin\theta_c$ = $n_2 \sin 90°$ gives $\theta_c = \sin^{-1}(1/1.333)$ = $\sin^{-1}(0.750) = 48.6°$.



ANS. FIG. P35.53

The radius then satisfies

$$\tan\theta_c = \frac{r}{1.00\ \text{m}}$$

So the diameter is

$$d = 2\left[(1.00\ \text{m})\tan\theta_c\right]$$

$$d = (2.00\ \text{m})\tan 48.6° = \boxed{2.27\ \text{m}}$$

P35.54 If the light ray to the eyes of the scuba diver makes an angle of 38.0° with the horizontal, it makes an angle of 52.0° with the normal to the water surface. This is larger than the critical angle of 48.8° found in Example 35.6, however. Therefore, no light from above the water will approach the scuba diver's eyes from this direction. The light approaching from this direction will be that originating underwater and reflected downward from the surface. The Sun will be seen somewhere within a circle whose edge is 90.0° – 48.8° = 41.2° above the horizontal.

P35.55 From the textbook Figure P35.55, we have $w = 2b + a$, so

$$b = \frac{w-a}{2} = \frac{700\ \mu\text{m} - 1\ \mu\text{m}}{2} = 349.5\ \mu\text{m}$$

$$\tan\theta_2 = \frac{b}{t} = \frac{349.5\ \mu\text{m}}{1\,200\ \mu\text{m}} = 0.291 \quad \rightarrow \quad \theta_2 = 16.2°$$

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

For refraction at entry,

$$\theta_1 = \sin^{-1}\frac{n_2 \sin\theta_2}{n_1} = \sin^{-1}\left(\frac{1.55\sin 16.2°}{1.00}\right) = \sin^{-1} 0.433 = \boxed{25.7°}$$

P35.56 The incident light reaches the left-hand mirror at distance

$$(1.00\ \text{m})\tan 5.00° = 0.087\ 5\ \text{m}$$

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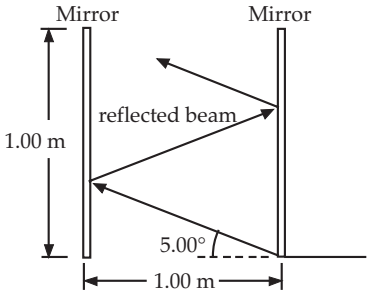
above its bottom edge. The reflected light first reaches the right-hand mirror at height

$$2(0.087\ 5\ \text{m}) = 0.175\ \text{m}$$

It bounces between the mirrors with this distance between points of contact with either. Since

$$\frac{1.00\ \text{m}}{0.175\ \text{m}} = 5.72$$

the light reflects



ANS. FIG. P35.56

five times from the right-hand mirror and six times from the left.

***P35.57** (a) The fraction reflected is

$$\frac{S_1'}{S_1} = \left[\frac{n_2 - n_1}{n_2 + n_1}\right]^2 = \left[\frac{1.52 - 1.00}{1.52 + 1.00}\right]^2 = \boxed{0.042\ 6}$$

(b) If medium 1 is glass and medium 2 is air,

$$\frac{S_1'}{S_1} = \left[\frac{n_2 - n_1}{n_2 + n_1}\right]^2 = \left[\frac{1.00 - 1.52}{1.00 + 1.52}\right]^2 = 0.042\ 6$$

There is no difference.

***P35.58** (a) With $n_1 = 1$ and $n_2 = n$, the reflected fractional intensity is

$$\frac{S_1'}{S_1} = \left(\frac{n-1}{n+1}\right)^2$$

The remaining intensity must be transmitted:

$$\frac{S_2}{S_1} = 1 - \left(\frac{n-1}{n+1}\right)^2 = \frac{(n+1)^2 - (n-1)^2}{(n+1)^2} = \frac{n^2 + 2n + 1 - n^2 + 2n - 1}{(n+1)^2}$$

$$= \boxed{\frac{4n}{(n+1)^2}}$$

(b) At entry, $\dfrac{S_2}{S_1} = \dfrac{4n}{(n+1)^2} = \dfrac{4(2.419)}{(2.419+1)^2} = 0.828.$

At exit, $\dfrac{S_3}{S_2} = 0.828.$

Overall, $\dfrac{S_3}{S_1} = \left(\dfrac{S_3}{S_2}\right)\left(\dfrac{S_2}{S_1}\right) = (0.828)^2 = 0.685$

or 68.5%.

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P35.59 Let $n(x)$ be the index of refraction at distance x below the top of the atmosphere and $n(x = h) = 1.00\ 293$ be its value at Earth's surface. Then,

$$n(x) = 1.000\ 00 + \left(\frac{1.002\ 93 - 1.000\ 00}{h}\right)x$$

$$= 1.000\ 00 + \left(\frac{0.002\ 93}{h}\right)x$$

(a) The total time interval required to traverse the atmosphere is

$$\Delta t = \int_0^h \frac{dx}{v} = \int_0^h \frac{n(x)}{c}dx: \quad \Delta t = \frac{1}{c}\int_0^h \left[1.000\ 00 + \left(\frac{0.002\ 93}{h}\right)x\right]dx$$

$$\Delta t = \frac{h}{c} + \frac{0.002\ 93}{ch}\left(\frac{h^2}{2}\right)$$

$$= \frac{h}{c}\left(\frac{2.002\ 93}{2}\right) = \frac{100 \times 10^3\ \text{m}}{3.00 \times 10^8\ \text{m/s}}\left(\frac{2.002\ 93}{2}\right)$$

$$= 3.33 \times 10^{-4}\ \text{s} = \boxed{334\ \mu\text{s}}$$

(b) The travel time in the absence of an atmosphere would be $\frac{h}{c}$.

Thus, the time in the presence of an atmosphere is

$$\frac{h/c\left(\frac{2.002\ 93}{2}\right) - h/c}{h/c} = \left(\frac{0.002\ 93}{2}\right) \times 100\% = \boxed{0.147\%}$$

P35.60 Let $n(x)$ be the index of refraction at distance x below the top of the atmosphere and $n(x = h) = n$ be its value at the planet surface.

Then, $$n(x) = 1.00 + \left(\frac{n - 1.00}{h}\right)x$$

(a) The total time interval required to traverse the atmosphere is

$$\Delta t = \int_0^h \frac{dx}{v} = \int_0^h \frac{n(x)}{c}dx : \Delta t = \frac{1}{c}\int_0^h \left[1.00 + \left(\frac{n - 1.00}{h}\right)x\right]dx$$

$$\Delta t = \frac{h}{c} + \frac{(n - 1.00)}{ch}\left(\frac{h^2}{2}\right) = \boxed{\frac{h}{c}\left(\frac{n + 1.00}{2}\right)}$$

(b) The travel time in the absence of an atmosphere would be $\frac{h}{c}$.

Thus, the time in the presence of an atmosphere is

$$\boxed{\left(\frac{n + 1.00}{2}\right)\ \text{times larger}}$$

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P35.61 Let the air and glass be medium 1 and 2, respectively. By Snell's law,

$$n_2 \sin\theta_2 = n_1 \sin\theta_1$$

or $$1.56 \sin\theta_2 = \sin\theta_1$$

But the conditions of the problem are such that $\theta_1 = 2\theta_2$, so $1.56\sin\theta_2 = \sin 2\theta_2$ We now use the double-angle trig identity suggested:

$$1.56 \sin\theta_2 = 2\sin\theta_2 \cos\theta_2$$

or $$\cos\theta_2 = \frac{1.56}{2} = 0.780$$

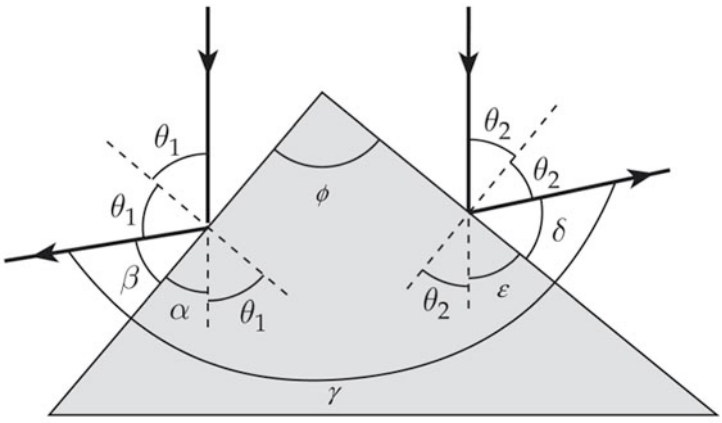
Thus, $\theta_2 = 38.7°$ and $\theta_1 = 2\theta_2 = \boxed{77.5°}$.

P35.62 In ANS. FIG. P35.62, observe on the left side of the prism that $\beta = 90° - \theta_1$ and $\alpha = 90° - \theta_1$. Thus, $\beta = \alpha$. Similarly, on the right side of the prism, $\delta = 90° - \theta_2$ and $\varepsilon = 90° - \theta_2$, giving $\delta = \varepsilon$. The incident rays are initially parallel, so observe that the angle between the reflected rays is $\gamma = (\alpha + \beta) + (\varepsilon + \delta)$, so $\gamma = 2(\alpha + \varepsilon)$. Finally, observe that the left side of the prism is sloped at angle α from the vertical, and the right side is sloped at angle ε. The angle ϕ is related to the other angles by

$$\phi + (90° - \alpha) + (90° - \varepsilon) = 180° \;\rightarrow\; \phi = \alpha + \varepsilon$$

Thus, we obtain the result

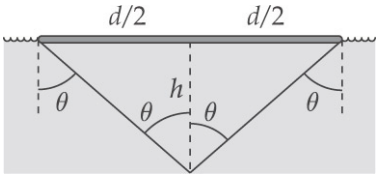
$$\gamma = 2(\alpha + \varepsilon) = 2\phi \;\rightarrow\; \phi = \frac{1}{2}\gamma$$



ANS. FIG. P35.62

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P35.63 Light from the diamond reflects totally at the water's surface at incident angles greater than the critical angle θ_c. The circular raft must cover the area of the surface through which light from the diamond could emerge. Thus, it must form the base of an inverted cone (with apex at the diamond) whose half angle is at least the critical angle.



ANS. FIG. P35.63

$$\theta \geq \theta_c$$

$$\tan\theta \geq \tan\theta_c$$

$$\frac{d/2}{h} \geq \tan\theta_c \;\rightarrow\; h \leq \frac{d}{2\tan\theta_c}$$

The critical angle at the water-air boundary is

$$\theta_c = \sin^{-1}\left(\frac{n_{\text{air}}}{n_{\text{water}}}\right) = \sin^{-1}\left(\frac{1.000}{1.333}\right) = 48.61°$$

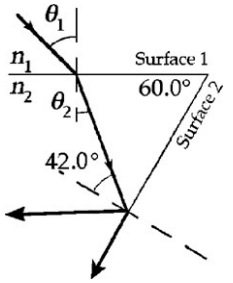
Thus, the maximum depth of the water is

$$h_{\max} = \frac{d}{2\tan\theta_c} = \frac{4.54\text{ m}}{2\tan 48.61°} = \boxed{2.00\text{ m}}$$

***P35.64** Consider an insulated box with the imagined one-way mirror forming one face, installed so that 90% of the electromagnetic radiation incident from the outside is transmitted to the inside and only a lower percentage of the electromagnetic waves from the inside make it through to the outside. Suppose the interior and exterior of the box are originally at the same temperature. Objects within and without are radiating and absorbing electromagnetic waves. They would all maintain constant temperature if the box had an open window. With the glass letting more energy in than out, the interior of the box will rise in temperature. But this is impossible, according to Clausius's statement of the second law. This reduction to a contradiction proves that it is impossible for the one-way mirror to exist.

P35.65 Define n_1 to be the index of refraction of the surrounding medium and n_2 to be that for the prism material. We can use the critical angle of 42.0° to find the ratio $\frac{n_2}{n_1}$:

$$n_2 \sin 42.0° = n_1 \sin 90.0°$$



ANS. FIG. P35.65

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So, $$\frac{n_2}{n_1} = \frac{1}{\sin 42.0^\circ} = 1.49$$

Call the angle of refraction θ_2 at the surface 1. The ray inside the prism forms a triangle with surfaces 1 and 2, so the sum of the interior angles of this triangle must be 180°.

Thus, $$(90.0^\circ - \theta_2) + 60.0^\circ + (90.0^\circ - 42.0^\circ) = 180^\circ$$

Therefore, $\theta_2 = 18.0^\circ$.

Applying Snell's law at surface 1, $n_1 \sin\theta_1 = n_2 \sin 18.0^\circ$:

$$\sin\theta_1 = \left(\frac{n_2}{n_1}\right)\sin\theta_2 = 1.49 \sin 18.0^\circ$$

gives $\boxed{\theta_1 = 27.5^\circ}$

P35.66 The number N of reflections the beam makes before exiting at the other end is equal to the length of the slab divided by the component of the displacement of the beam for each reflection:

$$N = \frac{L}{(t / \tan\theta_2)} = \frac{L \tan\theta_2}{t}$$

where θ_2 is the refracted angle as the beam enters the material. Substitute for this refracted angle in terms of the incident angle by using Snell's law:

$$N = \frac{L}{t}\tan\left[\sin^{-1}\left(\frac{n_1 \sin\theta_1}{n_2}\right)\right]$$

Substitute numerical values:

$$N = \frac{0.420 \text{ m}}{0.003\ 10 \text{ m}}\tan\left[\sin^{-1}\left(\frac{(1)\sin 50.0^\circ}{1.48}\right)\right]$$
$$= 81.96 \rightarrow 81 \text{ reflections}$$

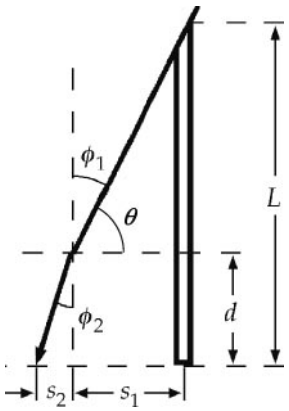
Therefore, the beam will exit after making 81 reflections, so it does not make 85 reflections.

P35.67 A light beam passing the top of the pole makes an angle θ of 40.0° with the horizontal, so its angle of incidence at the water is $\phi_1 = 90.0^\circ - \theta$. It enters the water's surface at distance from the pole

$$s_1 = \frac{L - d}{\tan\theta}$$

and has an angle of refraction ϕ_2 from $1.00 \sin\phi_1 = n \sin\phi_2$.

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ANS. FIG. P35.67

The beam reaches the bottom after traveling the horizontal distance

$$s_2 = d\tan\phi_2$$

The whole shadow length is

$$s_1 + s_2 = \frac{L-d}{\tan\theta} + d\tan\left[\sin^{-1}\left(\frac{\sin\phi_1}{n}\right)\right]$$

Because $\sin\phi_1 = \sin(90.0° - \theta) = \cos\theta,$ we find that

$$s_1 + s_2 = \frac{L-d}{\tan\theta} + d\tan\left[\sin^{-1}\left(\frac{\cos\theta}{n}\right)\right]$$

$$= \frac{2.00\text{ m}}{\tan 40.0°} + (2.00\text{ m})\tan\left[\sin^{-1}\left(\frac{\cos 40.0°}{1.33}\right)\right] = \boxed{3.79\text{ m}}$$

P35.68 From Table 35.1, the index of refraction of polystyrene is 1.49.

(a) For polystyrene *surrounded by air*, total internal reflection requires

$$\theta_3 \geq \theta_c = \sin^{-1}\left(\frac{1.00}{1.49}\right) = 42.2°$$

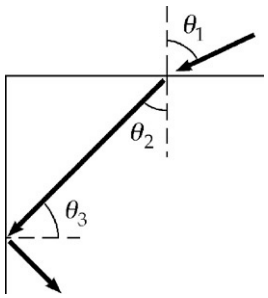
Then from geometry, $\theta_2 = 90.0° - \theta_3 \leq 47.8°.$

From Snell's law,

$$\sin\theta_1 = 1.49\sin\theta_2 \leq 1.49\sin 47.8°$$

$$\sin\theta_1 \leq 1.10$$

Any angle θ_1 satisfies this equation.



ANS. FIG. P35.68

Total internal reflection occurs for all values of θ, or the maximum angle is 90°.

(b) For polystyrene *surrounded by water*, $\theta_3 = \sin^{-1}\left(\frac{1.33}{1.49}\right) = 63.2°$

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and $\theta_2 = 26.8°$.

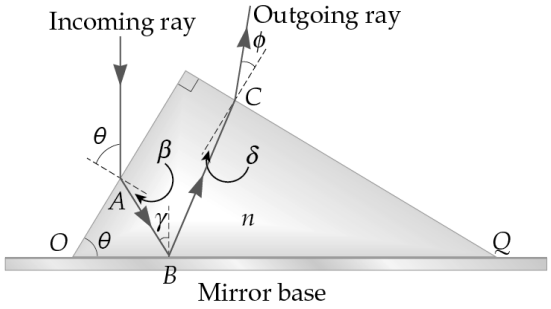
From Snell's law, $\theta_1 = \boxed{30.3°}$.

(c) From Table 35.1, the index of carbon disulfide is 1.628 > 1.49. Total internal reflection never occurs as the light moves from lower-index polystyrene into higher-index carbon disulfide.

P35.69 From ANS. FIG. P35.69, observe that the angle of incidence at *A* is the same as the prism angle at point *O*. Given that $\theta = 60.0°$, application of By Snell's law at point *A*:

$$1.50 \sin\beta = (1.00)\sin 60.0°$$

$$\sin\beta = \frac{\sin 60.0°}{1.50}$$



ANS. FIG. P35.69

From triangle *AOB*, we calculate the angle of incidence and reflection, γ, at point *B*:

$$\theta + (90.0° - \beta) + (90.0° - \gamma) = 180° \quad \text{or} \quad \gamma = \theta - \beta$$

Now, we find the angle of incidence at point *C* using triangle *BCQ*:

$$(90.0° - \gamma) + (90.0° - \delta) + (90.0° - \theta) = 180°$$

or

$$\delta = 90.0° - (\theta + \gamma) = 90.0° - (\theta + \theta - \beta) = 90.0° - 2\theta + \beta$$

Finally, application of Snell's law at point *C* gives

$$(1.00)\sin\phi = (1.50)\sin\delta$$

or

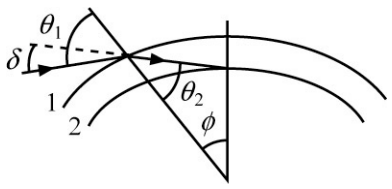
$$\phi = \sin^{-1}\left[1.50 \sin(90.0° - 2\theta + \beta)\right]$$

$$= \sin^{-1}\left\{1.50 \sin\left[90.0° - 2(60.0°) + \sin^{-1}\left(\frac{\sin 60.0°}{1.50}\right)\right]\right\}$$

$$= \boxed{7.91°}$$

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P35.70 (a) The optical day is longer. Incoming sunlight is refracted downward at the top of the atmosphere, so an observer can see the rising Sun when it is still geometrically below the horizon. Light from the setting Sun reaches her after the Sun is below the horizon geometrically.



ANS. FIG. P35.70

(b) ANS. FIG. P35.70 illustrates optical sunrise. At the center of the Earth,

$$\cos\phi = \frac{6.37\times10^6\ \text{m}}{6.37\times10^6\ \text{m} + 8\,614}$$

$$\phi = 2.98°$$

$$\theta_2 = 90 - 2.98° = 87.0°$$

At the top of the atmosphere

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

$$1\sin\theta_1 = 1.000\,293\sin 87.0°$$

$$\theta_1 = 87.4°$$

Deviation upon entry is

$$\delta = |\theta_1 - \theta_2|$$

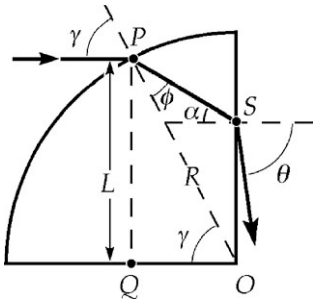
$$\delta = 87.364° - 87.022° = 0.342°$$

Sunrise of the optical day is before geometric sunrise by $0.342°\left(\frac{86\,400\ \text{s}}{360°}\right) = 82.2\ \text{s}$. Optical sunset occurs later too, so the optical day is longer by $\boxed{164\ \text{s}}$.

P35.71 Observe in ANS. FIG. P35.71 that the angle of incidence at point P is γ, and using triangle OPQ:

$$\sin\gamma = \frac{L}{R}$$

Also, $$\cos\gamma = \sqrt{1-\sin^2\gamma} = \frac{\sqrt{R^2 - L^2}}{R}$$



ANS. FIG. P35.71

Apply Snell's law at point P:

$$1.00\sin\gamma = n\sin\phi$$

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Thus, $\sin\phi = \dfrac{\sin\gamma}{n} = \dfrac{L}{nR}$

and $\cos\phi = \sqrt{1-\sin^2\phi} = \dfrac{\sqrt{n^2R^2 - L^2}}{nR}.$

From triangle OPS, $\phi + (\alpha + 90.0°) + (90.0° - \gamma) = 180°,$ or the angle of incidence at point S is $\alpha = \gamma - \phi$. Then, applying Snell's law at point S gives $1.00\sin\theta = n\sin\alpha = n\sin(\gamma - \phi)$

or
$$\begin{aligned}\sin\theta &= n\sin(\gamma-\phi)\\ &= n[\sin\gamma\cos\phi - \cos\gamma\sin\phi]\\ &= n\left[\left(\frac{L}{R}\right)\frac{\sqrt{n^2R^2-L^2}}{nR} - \frac{\sqrt{R^2-L^2}}{R}\left(\frac{L}{nR}\right)\right]\\ &= \frac{L}{R^2}\left(\sqrt{n^2R^2-L^2} - \sqrt{R^2-L^2}\right)\end{aligned}$$

thus, $\theta = \boxed{\sin^{-1}\left[\dfrac{L}{R^2}\left(\sqrt{n^2R^2-L^2} - \sqrt{R^2-L^2}\right)\right]};$

or, using from above $\sin\gamma = \dfrac{L}{R} \rightarrow \gamma = \sin^{-1}\dfrac{L}{R}$ and $\phi = \sin^{-1}\dfrac{L}{nR}$,

$$\sin\theta = n\sin(\gamma-\phi) = n\sin\left(\sin^{-1}\frac{L}{R} - \sin^{-1}\frac{L}{nR}\right)$$

$$\theta = \boxed{\sin^{-1}\left[n\sin\left(\sin^{-1}\frac{L}{R} - \sin^{-1}\frac{L}{nR}\right)\right]}$$

P35.72 $\delta = \theta_1 - \theta_2 = 10.0°$ and $n_1\sin\theta_1 = n_2\sin\theta_2$ with $n_1 = 1,\ n_2 = \dfrac{4}{3}.$

Thus, $\theta_1 = \sin^{-1}(n_2\sin\theta_2) = \sin^{-1}[n_2\sin(\theta_1 - 10.0°)].$

(You can use a calculator to home in on an approximate solution to this equation, testing different values of θ_1 until you find that $\theta_1 = \boxed{36.5°}$. Alternatively, you can solve for θ_1 exactly, as shown below.)

We are given that $\sin\theta_1 = \dfrac{4}{3}\sin(\theta_1 - 10.0°).$

This is the sine of a difference, so

$$\frac{3}{4}\sin\theta_1 = \sin\theta_1\cos 10.0° - \cos\theta_1\sin 10.0°$$

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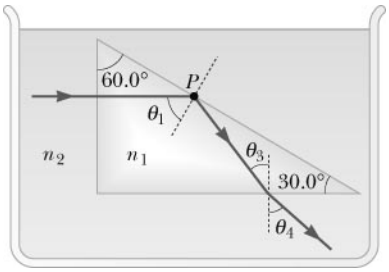
Rearranging, $\sin 10.0^\circ \cos\theta_1 = \left(\cos 10.0^\circ - \frac{3}{4}\right)\sin\theta_1$,

$$\frac{\sin 10.0^\circ}{\cos 10.0^\circ - 0.750} = \tan\theta_1$$

and $\theta_1 = \tan^{-1}(0.740) = \boxed{36.5^\circ}$.

P35.73 (a) From the geometry shown in ANS. FIG. P35.73, observe that $\theta_1 = 60.0^\circ$. Also, from the law of reflection, $\theta_2 = \theta_1 = 60.0^\circ$.

Therefore, $\alpha = 90.0^\circ - \theta_2 = 30.0^\circ$, and $\theta_3 + 90.0^\circ = 180 - \alpha - 30.0^\circ$ or $\theta_3 = 30.0^\circ$.



ANS. FIG. P35.73

Then, since the prism is immersed in water ($n_2 = 1.333$), Snell's law gives

$$\theta_4 = \sin^{-1}\left(\frac{n_{\text{glass}}\sin\theta_3}{n_2}\right) = \sin^{-1}\left(\frac{(1.66)\sin 30.0^\circ}{1.333}\right) = \boxed{38.5^\circ}$$

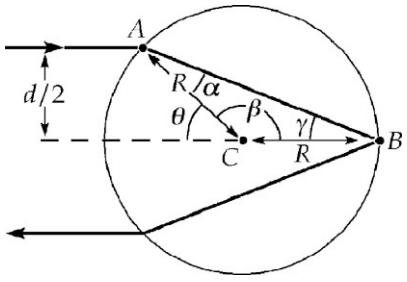
(b) For refraction to occur at point P, it is necessary that $\theta_c > \theta_1$. Thus,

$\theta_c = \sin^{-1}\left(\frac{n_2}{n_{\text{glass}}}\right) > \theta_1$, which gives

$n_2 > n_{\text{glass}}\sin\theta_1 = (1.66)\sin 60.0^\circ = \boxed{1.44}$

P35.74 As shown in ANS. FIG. P35.74, the angle of incidence at point A is:

$$\theta = \sin^{-1}\left(\frac{d/2}{R}\right) = \sin^{-1}\left(\frac{1.00\text{ m}}{2.00\text{ m}}\right) = 30.0^\circ$$



ANS. FIG. P35.74

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If the emerging ray is to be parallel to the incident ray, the path must be symmetric about the centerline *CB* of the cylinder. In the isosceles triangle *ABC*,

$$\gamma = \alpha \quad \text{and} \quad \beta = 180° - \theta$$

Therefore, $\alpha + \beta + \gamma = 180°$

becomes $2\alpha + 180° - \theta = 180°$

or $$\alpha = \frac{\theta}{2} = 15.0°.$$

Then, applying Snell's law at point *A*,

$$n \sin\alpha = 1.00 \sin\theta$$

$$n = \frac{\sin\theta}{\sin\alpha} = \frac{\sin 30.0°}{\sin 15.0°} = \boxed{1.93}$$

P35.75 Applying Snell's law at points A, B, and C gives

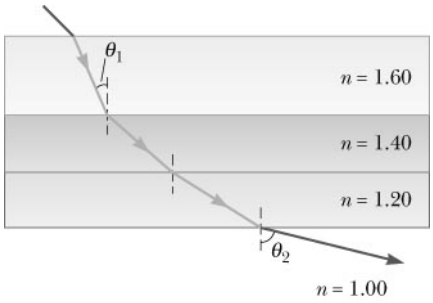
$$1.40 \sin\alpha = 1.60 \sin\theta_1 \qquad \textbf{[1]}$$

$$1.20 \sin\beta = 1.40 \sin\alpha \qquad \textbf{[2]}$$

and $$1.00 \sin\theta_2 = 1.20 \sin\beta \qquad \textbf{[3]}$$

Combining equations [1], [2], and [3] yields

$$\sin\theta_2 = 1.60 \sin\theta_1 \qquad \textbf{[4]}$$



ANS. FIG. P35.75

Note that equation [4] is exactly what Snell's law would yield if the second and third layers of this "sandwich" were ignored. This will always be true if the surfaces of all the layers are parallel to each other.

(a) If $\theta_1 = 30.0°$, then equation [4] gives

$$\theta_2 = \sin^{-1}(1.60 \sin 30.0°) = \boxed{53.1°}$$

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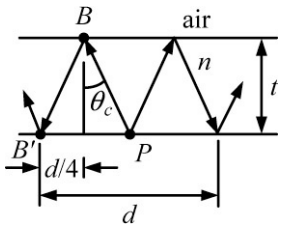
(b) At the critical angle of incidence on the lowest surface, $\theta_2 = 90.0°$. Then, equation [4] gives

$$\theta_1 = \sin^{-1}\left(\frac{\sin\theta_2}{1.60}\right) = \sin^{-1}\left(\frac{\sin 90.0°}{1.60}\right) = 38.7°$$

Total internal reflection will occur for $\boxed{\theta_1 \geq 38.7°}$.

P35.76 (a) At the boundary of the air and glass, the critical angle is given by

$$\sin\theta_c = \frac{1}{n}$$



ANS. FIG. P35.76

Consider the critical ray *PBB*′:

$$\tan\theta_c = \frac{d/4}{t} \quad \text{or} \quad \frac{\sin\theta_c}{\cos\theta_c} = \frac{d}{4t}$$

Squaring the last equation gives:

$$\frac{\sin^2\theta_c}{\cos^2\theta_c} = \frac{\sin^2\theta_c}{1-\sin^2\theta_c} = \left(\frac{d}{4t}\right)^2$$

Since $\sin\theta_c = \frac{1}{n}$, this becomes $\frac{1}{n^2-1} = \left(\frac{d}{4t}\right)^2$ or

$$\boxed{n = \sqrt{1+\left(\frac{4t}{d}\right)^2}}$$

(b) Solving for d,

$$d = \frac{4t}{\sqrt{n^2-1}}$$

Thus, if $n = 1.52$ and $t = 0.600$ cm, $d = \frac{4(0.600\text{ cm})}{\sqrt{(1.52)^2-1}} = \boxed{2.10\text{ cm}}$

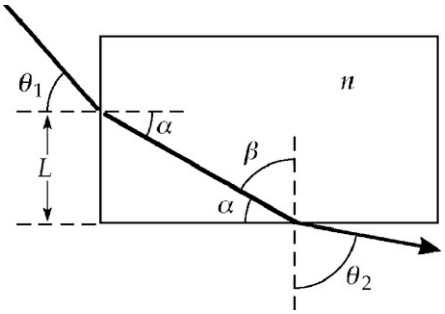
(c) Since violet light has a larger index of refraction, it will lead to a smaller critical angle and the inner edge of the white halo will be tinged with $\boxed{\text{violet}}$ light.

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P35.77 (a) Given that $\theta_1 = 45.0°$ and $\theta_2 = 76.0°$,

Snell's law at the first surface gives

$$n \sin\alpha = 1.00 \sin 45.0° \qquad \textbf{[1]}$$



ANS. FIG. P35.77

Observe that the angle of incidence at the second surface is

$$\beta = 90.0° - \alpha$$

Thus, Snell's law at the second surface yields

$$n \sin\beta = n \sin(90.0° - \alpha) = 1.00 \sin 76.0°$$

or $n \cos\alpha = \sin 76.0°$. **[2]**

Dividing equation [1] by equation [2], we obtain

$$\tan\alpha = \frac{\sin 45.0°}{\sin 76.0°} = 0.729$$

or $\alpha = 36.1°$.

Then, from equation [1],

$$n = \frac{\sin 45.0°}{\sin\alpha} = \frac{\sin 45.0°}{\sin 36.1°} = \boxed{1.20}$$

(b) From the sketch, observe that the distance the light travels in the plastic is $d = \dfrac{L}{\sin\alpha}$. Also, the speed of light in the plastic is $v = \dfrac{c}{n}$, so the time required to travel through the plastic is

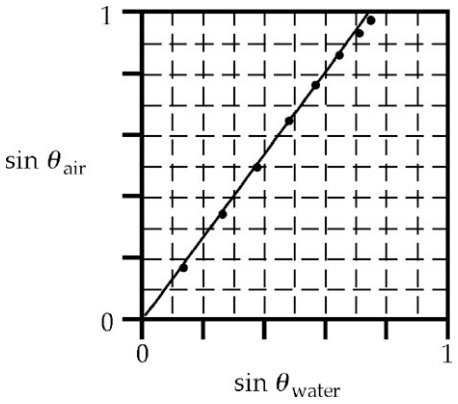
$$\Delta t = \frac{d}{v} = \frac{nL}{c \sin\alpha} = \frac{1.20(0.500 \text{ m})}{(3.00 \times 10^8 \text{ m/s}) \sin 36.1°}$$

$$= 3.40 \times 10^{-9} \text{ s} = \boxed{3.40 \text{ ns}}$$

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P35.78 (a) See graph in ANS. FIG. P35.78.

$\sin\theta_1$	$\sin\theta_2$	$\frac{\sin\theta_1}{\sin\theta_2}$
0.174	0.131	1.330 4
0.342	0.261	1.312 9
0.500	0.379	1.317 7
0.643	0.480	1.338 5
0.766	0.576	1.328 9
0.866	0.647	1.339 0
0.940	0.711	1.322 0
0.985	0.740	1.331 5



ANS. FIG. P35.78

(b) The straightness of the graph line demonstrates Snell's proportionality of the sine of the angle of refraction to the sine of the angle of incidence.

(c) The slope of the line is $\overline{n} = 1.327\,6 \pm 0.01$

The equation $\sin\theta_1 = n\sin\theta_2$ shows that this slope is the index of refraction, $n = \boxed{1.328 \pm 0.8\%}$

P35.79 (a) We see the Sun moving from east to west across the sky. Its angular speed is

$$\omega = \frac{\Delta\theta}{\Delta t} = \frac{2\pi \text{ rad}}{86\,400 \text{ s}} = 7.27 \times 10^{-5} \text{ rad/s}$$

The direction of sunlight crossing the cell from the window changes at this rate, moving on the opposite wall at speed

$$v = r\omega = (2.37 \text{ m})\left(7.27 \times 10^{-5} \text{ rad/s}\right)$$
$$= 1.72 \times 10^{-4} \text{ m/s} = \boxed{0.172 \text{ mm/s}}$$

(b) The mirror folds into the cell the motion that would occur in a room twice as wide:

$$v = r\omega = 2(0.174 \text{ mm/s}) = \boxed{0.345 \text{ mm/s}}$$

(c), (d) As the Sun moves southward and upward at 50.0°, we may regard the corner of the window as fixed, and both patches of light move $\boxed{\text{northward and downward at } 50.0°}$.

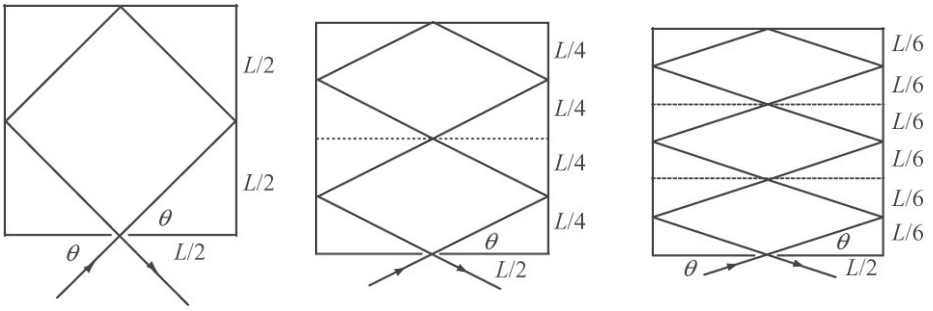
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P35.80 Because the enclosure is square and the beam enters at bottom center, and because a light beam travels the same path regardless of its direction on the path, we expect the beam pattern to be symmetric about a vertical line passing through the opening. Therefore, the beam enters the opening at the same angle it exits, the beam strikes each side mirror at the same height, and the beam forms a zigzag pattern that intersects itself at a point (or points) above the center opening; thus, the beam must reflect off the top mirror at its center. Also, because of the law of reflection, the path of the beam is symmetric about a horizontal line passing through the points where the beam reflects off a side mirror.

(a) Call the length of each side of the square L. If the beam is to strike each mirror once, the beam must strike each side mirror at its center, at height $L/2$ after traveling a horizontal distance $L/2$. Therefore,

$$\tan\theta = \frac{L/2}{L/2} = 1 \;\rightarrow\; \theta = 45.0^\circ$$

The beam will exit the enclosure if it enters at angle $\boxed{45.0^\circ}$, as shown in ANS. FIG. P35.80(a).



ANS. FIG. P35.80(a) **ANS. FIG. P35.80(b)** **ANS. FIG. P35.80(c)**

(b) Because the path of the beam is symmetric about a horizontal lines passing through the points where the beam reflects off a side mirror, we can divide the square enclosure into vertically stacked rectangular areas, each a mirror image of the one below. In each, the ray passes upward through the bottom center of the rectangle and exits at its top center until it reflects off the top mirror, then the ray passes back downward through each center until it exits the enclosure. The pattern of the ray's path is repeated in each rectangle. If the enclosure is divided into n rectangles, the height of each rectangle is L/n, and the beam strikes a side mirror at height $L/2n$ within each rectangle. Therefore, the angle of entry at the opening is

$$\tan\theta = \frac{L/2n}{L/2} = \frac{1}{n}$$

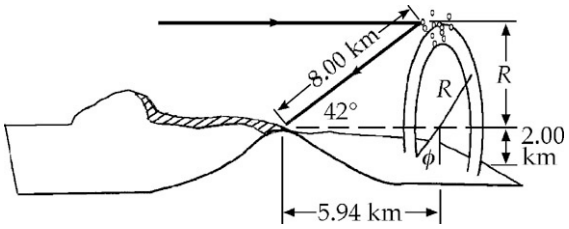
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The cases for $n = 2$ and 3 are shown in ANS. FIG. 35.80(b) and (c) above.

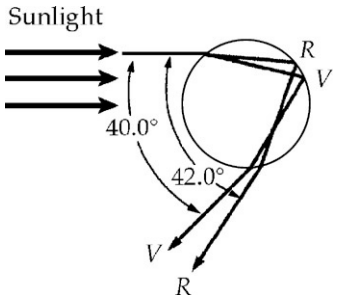
Yes. The ray will exit if it enters at an angle θ that satisfies the condition $\tan\theta = \dfrac{1}{n}$, where $n = 1,\ 2,\ 3,\ \ldots$

Challenge Problems

P35.81 Horizontal light rays from the setting Sun pass above the hiker. The light rays are twice refracted and once reflected, as in ANS. FIG. P35.81(b). The most intense light reaching the hiker, that which represents the visible rainbow, is located between angles of 40° and 42° from the hiker's shadow.



(a)



(b)

ANS. FIG. P35.81

The hiker sees a greater percentage of the violet inner edge, so we consider the red outer edge. The radius R of the circle of droplets is

$$R = (8.00\ \text{km})\sin 42.0° = 5.35\ \text{km}$$

Then the angle ϕ, between the vertical and the radius where the bow touches the ground, is given by

$$\cos\phi = \frac{2.00\ \text{km}}{R} = \frac{2.00\ \text{km}}{5.35\ \text{km}} = 0.374$$

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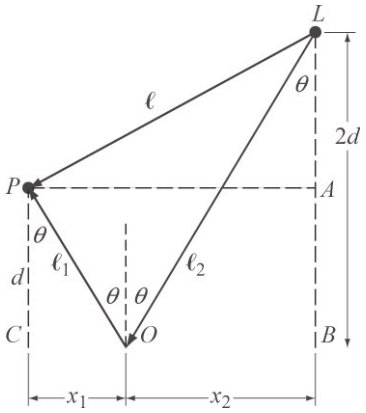
or $\phi = 68.1°$.

The angle filled by the visible bow is

$$360° - (2 \times 68.1°) = 224°$$

so the visible bow is $\frac{224°}{360°} = \boxed{62.2\% \text{ of a circle}}$.

P35.82 The geometry of the situation is shown in ANS. FIG. P35.82, where P is the person and L is the lightbulb.



ANS. FIG. P35.82

We have used the law of reflection to claim that the angles on either side of the dashed line at O are equal. From triangle OPC, we see that

$$\cos\theta = \frac{d}{\ell_1} \quad \text{and} \quad \sin\theta = \frac{x_1}{\ell_1}$$

which can be rearranged to give

$$\ell_1 = \frac{d}{\cos\theta} \quad \text{and} \quad x_1 = \ell_1 \sin\theta \qquad \textbf{[1]}$$

Similarly, from triangle OLB,

$$\cos\theta = \frac{2d}{\ell_2} \quad \text{and} \quad \sin\theta = \frac{x_2}{\ell_2}$$

which can be rearranged to give

$$\ell_2 = \frac{2d}{\cos\theta} \quad \text{and} \quad x_2 = \ell_2 \sin\theta \qquad \textbf{[2]}$$

Let $n = 3.10$ from the problem statement. The condition given in the problem is expressed as

$$\ell_1 + \ell_2 = n\ell \qquad \textbf{[3]}$$

Substitute for ℓ_1 and ℓ_2 from equations [1] and [2]:

$$\frac{d}{\cos\theta} + \frac{2d}{\cos\theta} = n\ell \rightarrow \frac{3d}{\cos\theta} = n\ell \qquad \textbf{[4]}$$

From triangle APL, apply the Pythagorean theorem:

$$\ell^2 = d^2 + (x_1 + x_2)^2$$

Substitute for x_1 and x_2 from equations [1] and [2]:

$$\ell^2 = d^2 + (\ell_1 \sin\theta + \ell_2 \sin\theta)^2 = d^2 + (\ell_1 + \ell_2)^2 \sin^2\theta$$

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Substitute from equation [3]:

$$\ell^2 = d^2 + n^2\ell^2 \sin^2\theta \;\rightarrow\; \ell^2\left(1 - n^2\sin^2\theta\right) = d^2 \qquad \textbf{[5]}$$

Eliminate ℓ between equations [4] and [5]:

$$\left(\frac{3d}{n\cos\theta}\right)^2\left(1 - n^2\sin^2\theta\right) = d^2 \;\rightarrow\; 9 - 9n^2\sin^2\theta = n^2\cos^2\theta$$

Simplify this expression:

$$9 = 9n^2\sin^2\theta + n^2\cos^2\theta = 8n^2\sin^2\theta + n^2\sin^2\theta + n^2\cos^2\theta$$

$$= 8n^2\sin^2\theta + n^2 \;\rightarrow\; \sin\theta = \sqrt{\frac{9 - n^2}{8n^2}}$$

If we now substitute $n = 3.10$, we see that there is no real solution for $\sin\theta$. Therefore, it is impossible for the distances to be in this relationship. The largest value that n can have is 3.00, which leads to an incident angle of 0°.

In fact, we could have solved this problem more elegantly (and quickly!) by realizing that the largest ratio of distances would be obtained by bringing the person and the lightbulb as close together as possible given the condition on their distances from the mirror. This would be done by aligning them both above O in the figure so that the light strikes the mirror at normal incidence. Then, the person and lightbulb are separated by a distance d, and the light travels a distance $3d$. This gives a maximum ratio of 3.00 and we see that a ratio of 3.10 is impossible.

P35.83 (a) Calling the angle between the dashed line in Figure P35.83 and the reflected laser beam θ, we see that

$$\tan\theta = \frac{x}{L/2} = \frac{2x}{L} \;\rightarrow\; x = \frac{1}{2}L\tan\theta$$

Differentiate with respect to time to find the speed of the laser spot on the wall:

$$v = \frac{dx}{dt} = \frac{d}{dt}\left(\frac{1}{2}L\tan\theta\right) = \frac{1}{2}L\sec^2\theta\frac{d\theta}{dt} \qquad \textbf{[1]}$$

From Figure P35.83, we see that

$$\sec\theta = \frac{1}{\cos\theta} = \frac{\sqrt{4x^2 + L^2}}{L} \qquad \textbf{[2]}$$

Because the incident ray is stationary, as the mirror turns through angle ϕ, its normal rotates through angle ϕ, so the angle of

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incidence increases by ϕ as does the angle of reflection. Therefore, the reflected ray rotates through 2ϕ. As a consequence, the angular speed of the reflected ray is twice that of the mirror:

$$\omega_{\text{reflected ray}} = \frac{d\theta}{dt} = 2\omega \qquad \textbf{[3]}$$

Substitute equations [2] and [3] into equation [1]:

$$v = \frac{1}{2}L\left(\frac{4x^2 + L^2}{L^2}\right)2\omega = \boxed{\left(\frac{4x^2 + L^2}{L}\right)\omega}$$

(b) The variable in this expression is x, so we can minimize the speed by setting $x = \boxed{0}$.

(c) Let $x = 0$ in the expression for v:

$$v = \left(\frac{4(0)^2 + L^2}{L}\right)\omega = \boxed{L\omega}$$

(d) The maximum speed occurs when the reflected laser beam arrives at a corner of the room, where $x = L/2$:

$$v = \left(\frac{4(L/2)^2 + L^2}{L}\right)\omega = \boxed{2L\omega}$$

(e) Between the minimum and maximum speed, the reflected laser beam rotates through $\pi/4$ radians, so the mirror rotates through $\pi/8$ radians. Therefore,

$$\Delta t = \frac{\Delta\theta}{\omega} = \boxed{\frac{\pi}{8\omega}}$$

P35.84 (a) In the textbook Figure P35.84, we have $r_1 = \sqrt{a^2 + x^2}$ and $r_2 = \sqrt{b^2 + (d-x)^2}$. The speeds in the two media are $v_1 = c/n_1$ and $v_2 = c/n_2$ so the travel time for the light from P to Q is indeed

$$\Delta t = \frac{r_1}{v_1} + \frac{r_2}{v_2} = \frac{n_1\sqrt{a^2 + x^2}}{c} + \frac{n_2\sqrt{b^2 + (d-x)^2}}{c}$$

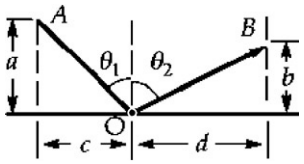
(b) Now $\dfrac{d(\Delta t)}{dx} = \dfrac{n_1}{2c}\dfrac{2x}{\sqrt{a^2 + x^2}} + \dfrac{n_2}{2c}\dfrac{2(d-x)(-1)}{\sqrt{b^2 + (d-x)^2}} = 0$ is the requirement for minimal travel time, which simplifies to

$$\frac{n_1 x}{\sqrt{a^2 + x^2}} = \frac{n_2(d-x)}{\sqrt{b^2 + (d-x)^2}}$$

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(c) Now $\sin\theta_1 = \dfrac{x}{\sqrt{a^2 + x^2}}$ and $\sin\theta_2 = \dfrac{d - x}{\sqrt{b^2 + (d - x)^2}}$, so we have $n_1 \sin\theta_1 = n_2 \sin\theta_2$.

P35.85 In ANS. FIG. P35.85, a ray travels along path *AM* from point *A* to the mirror, reflects and travels along path *MB* from the mirror to point *B*. Point *A* is a vertical distance *a* above the mirror, and point *B* is a vertical distance *b* above the mirror. Points *A* and *B* are a horizontal distance *d* apart. The ray strikes the mirror at point *M* which is a horizontal distance *x* from point *A*. The angle of incidence is θ_1 and the angle of reflection is θ_2.



ANS. FIG. P35.85

We have $AM = \sqrt{a^2 + x^2}$ and $MB = \sqrt{b^2 + (d - x)^2}$. The travel time for the light from *A* to *B* is

$$\Delta t = \frac{AM}{c} + \frac{MB}{c} = \frac{\sqrt{a^2 + x^2}}{c} + \frac{\sqrt{b^2 + (d - x)^2}}{c}$$

We require a minimal travel time, so

$$\frac{d(\Delta t)}{dx} = \frac{1}{2c}\frac{2x}{\sqrt{a^2 + x^2}} + \frac{1}{2c}\frac{2(d - x)(-1)}{\sqrt{b^2 + (d - x)^2}} = 0$$

which simplifies to

$$\frac{x}{\sqrt{a^2 + x^2}} = \frac{(d - x)}{\sqrt{b^2 + (d - x)^2}}$$

This expression is equivalent to

$$\sin\theta_1 = \sin\theta_2 \;\rightarrow\; \theta_1 = \theta_2$$

P35.86 (a) Assume the viewer is far away to the right. In ANS. FIG. P35.86(a), a ray directed toward the viewer comes tangentially from the edge of the glowing sphere and emerges from the atmosphere at angle θ_2. The apparent radius of the glowing sphere is R_3 as shown. For the figure, we see that

$$\sin\theta_1 = \frac{R_1}{R_2} \quad \text{and} \quad \sin\theta_2 = \frac{R_3}{R_2}$$

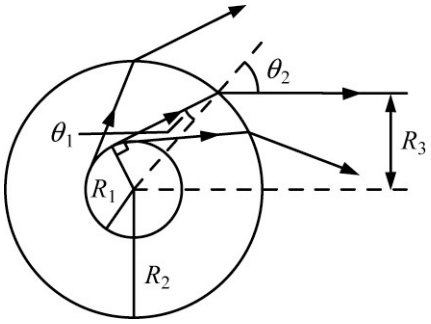
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Then,

$$n\sin\theta_1 = 1.00\sin\theta_2$$

and

$$n\frac{R_1}{R_2} = \frac{R_3}{R_2} \quad \rightarrow \quad \boxed{R_3 = nR_1}$$



ANS. FIG. P35.86(a)

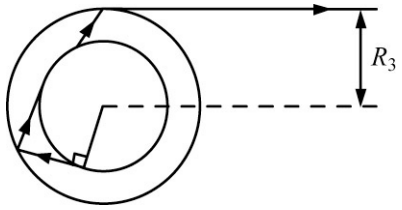
(b) If a ray is to come tangentially from the edge of the glowing sphere and emerge from the atmosphere, the incident angle θ_1 must be less than the critical angle, $\theta_1 < \theta_c$. Then,

$$\sin\theta_1 < \sin\theta_c = \frac{1}{n}$$

and

$$\frac{R_1}{R_2} < \frac{1}{n} \quad \rightarrow \quad nR_1 < R_2 \quad \rightarrow \quad R_2 > nR_1$$

This is not so for the case we consider here.



ANS. FIG. P35.86(b)

Thus, the ray considered in part (a) undergoes total internal reflection. In this case a ray traveling toward the viewer must emerge tangentially from the atmosphere, as shown in ANS. FIG. P35.86(b), so the apparent radius of the glowing sphere is the same as the radius of the atmosphere: $\boxed{R_3 = R_2}$.

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***P35.87** Define $T = \dfrac{4n}{(n+1)^2}$ as the transmission coefficient for one encounter with an interface. For diamond and air, it is 0.828, as in Problem P35.58. As shown in ANS. FIG. P35.87, the total amount transmitted is

$$T^2 + T^2(1-T)^2 + T^2(1-T)^4$$
$$+T^2(1-T)^6 + \ldots + T^2(1-T)^{2n} + \ldots$$

We have $1 - T = 1 - 0.828 = 0.172,$ so the total transmission is

$$(0.828)^2\left[1 + (0.172)^2 + (0.172)^4 + (0.172)^6 + \ldots\right]$$

To sum this series, define

$$F = 1 + (0.172)^2 + (0.172)^4 + (0.172)^6 + \ldots$$

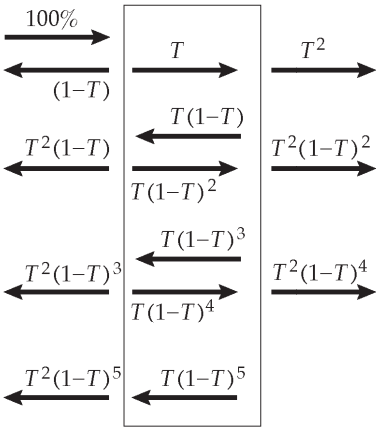
Note that $(0.172)^2 F = (0.172)^2 + (0.172)^4 + (0.172)^6 + \ldots,$ and

$$1 + (0.172)^2 F = 1 + (0.172)^2 + (0.172)^4 + (0.172)^6 + \ldots = F$$

Then,

$$1 = F - (0.172)^2 F \quad \text{or} \quad F = \frac{1}{1-(0.172)^2}.$$

The overall transmission is then $\dfrac{(0.828)^2}{1-(0.172)^2} = 0.706$ or $\boxed{70.6\%}$.



ANS. FIG. P35.87

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ANSWERS TO EVEN-NUMBERED PROBLEMS

P35.2 (a) 3.00×10^8 m/s; (b) The sizes of the objects need to be taken into account. Otherwise the answer would be too large by 2%.

P35.4 2.27×10^8 m/s

P35.6 19.5° above the horizon

P35.8 (a) $\theta = 30.4°$; (b) $\theta' = 22.3°$

P35.10 (a) See P35.10(a) for full explanation; (b) Now $CBE = \phi$ is the angle of incidence of the vertical light beam. Its angle of reflection is also ϕ. The angle between the vertical incident beam and the reflected beam is 2ϕ; (c) $\phi = 0.055\,7°$

P35.12 $\theta_2 = 19.5°$; $\theta_3 = 19.5°$; $\theta_4 = 30.0°$

P35.14 (a) 78.3°; (b) 2.56 m; (c) 9.72°; (d) 442 nm; (e) The light wave slows down as it moves from air to water, but the sound wave speeds up by a larger factor. The light wave bends toward the normal and its wavelength shortens, but the sound wave bends away from the normal and its wavelength increases.

P35.16 (a) 1.52; (b) 417 nm; (c) 4.74×10^{14} Hz; (d) 198 Mm/s

P35.18 $\beta = 180° - 2\theta$

P35.20 (a) See P35.20(a) for full explanation; (b) See P35.20(b) for full explanation.

P35.22 (a) 0.387 cm; (b) 106 ps

P35.24 (a) Yes, if the angle of incidence is 58.9°; (b) No. Both the reduction in speed and the bending toward the normal reduce the component of velocity parallel to the interface. This component cannot remain constant for a nonzero angle of incidence.

P35.26 6.30 cm

P35.28 (a) See P35.28(a) for full explanation; (b) 37.2°; (c) 37.3°; (d) 37.3°

P35.30 The index of refraction of the atmosphere decreases with increasing altitude because of the decrease in density of the atmosphere with increasing altitude. As indicated in the ray diagram, the Sun located at S below the horizon appears to be located at S′.

P35.32 (a) $\frac{h}{d} = \sqrt{\frac{n^2 - 1}{4 - n^2}}$; (b) 4.73 cm; (c) For $n = 1$, $h = 0$. For $n = 2$, $h = \infty$. For $n > 2$, h has no real solution.

P35.34 (a) See ANS. FIG. P35.34; (b) 42.0°; (c) 63.1°; (d) 26.9°; (e) 107 m

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P35.36 (a) 48.2°; (b) 47.8°

P35.38 (a) See ANS. FIG. P35.38(a); (b) As the waves move to shallower water, the wave fronts slow down, and those closer to shore slow down more. The rays tend to bend toward the normal of the contour lines; or equivalently, the wave fronts bend to become more nearly parallel to the contour lines; (c) See ANS. FIG. P35.38(c); (d) We suppose that the headlands are steep underwater, as they are above water. The rays are everywhere perpendicular to the wave fronts of the incoming refracting waves. As shown, because the rays tend to bend toward the normal of the contour lines, the rays bend toward the headlands and deliver more energy per length at the headlands.

P35.40 $$\sin^{-1}\left\{n_V \sin\left[\Phi - \sin^{-1}\left(\frac{\sin\theta}{n_V}\right)\right]\right\} - \sin^{-1}\left\{n_R \sin\left[\Phi - \sin^{-1}\left(\frac{\sin\theta}{n_R}\right)\right]\right\}$$

P35.42 (a) 27.0°; (b) 37.1°; (c) 49.8°

P35.44 $$\theta_1 > \sin^{-1}\left(n\sin\left[\Phi - \sin^{-1}\left(\frac{1.00}{n}\right)\right]\right);\ \theta_1 > \sin^{-1}\left(\sqrt{n^2-1}\sin\Phi - \cos\Phi\right)$$

P35.46 (a) 24.42°; (b) Because the angle of incidence (35.0°) is greater than the critical angle, the light is totally reflected at *P*; (c) 33.44°; (d) Yes. In this case, the angle of incidence is just larger than the critical angle, so the light ray again undergoes total internal reflection at *P*; (e) clockwise; (f) 2.83°

P35.48 (a) 10.7°; (b) air; (c) Sound in air falling on the wall from directions is 100% reflected.

P35.50 (a) See P35.50(a) for full explanation; (b) $n \geq 1.41$ and $n \leq 2.12$

P35.52 (a) angle of incidence: 30.0°, angle of refraction: 18.8°; (b) angle of incidence: 30.0°, angle of refraction: 50.8°; (c) and (d) See TABLE P35.52.

P35.54 No light from above the water will approach the scuba diver's eyes from 48.8° found in Example 35.6.

P35.56 Five times from the right-hand mirror and six times from the left.

P35.58 (a) $\frac{4n}{(n+1)^2}$; (b) 68.5%

P35.60 (a) $\frac{h}{c}\left(\frac{n+1.00}{2}\right)$; (b) $\left(\frac{n+1.00}{2}\right)$ times larger

P35.62 See P35.62 for full explanation.

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P35.64 Consider an insulated box with the imagined one-way mirror forming one face, installed so that 90% of the electromagnetic radiation incident from the outside is transmitted to the inside and only a lower percentage of the electromagnetic waves from the inside make it through to the outside. Suppose the interior and exterior of the box are originally at the same temperature. Objects within and without are radiating and absorbing electromagnetic waves. They would all maintain constant temperature if the box had an open window. With the glass letting more energy in than out, the interior of the box will rise in temperature. But this is impossible, according to Clausius's statement of the second law. This reduction to a contradiction proves that it is impossible for the one-way mirror to exist.

P35.66 The beam will exit after making 81 reflections, so it does not make 85 reflections.

P35.68 (a) Total internal reflection occurs for all values of θ, or the maximum angle is 90°; (b) 30.3°; (c) Total internal reflection never occurs as the light moves from lower-index polystyrene to higher-index carbon disulfide.

P35.70 (a) The optical day is longer; (b) 164 s

P35.72 36.5°

P35.74 1.93

P35.76 (a) $n = \sqrt{1 + \left(\frac{4t}{d}\right)^2}$; (b) 2.10 cm; (c) violet

P35.78 (a) See ANS. FIG. P35.78; (b) The straightness of the graph line demonstrates Snell's proportionality of the sine of the angle of refraction to the sine of the angle of incidence; (c) $1.328 \pm 0.8\%$

P35.80 (a) 45.0°; (b) Yes. The ray will exit if it enters at an angle θ that satisfies the condition $\tan\theta = \frac{1}{n}$, where n = 1, 2, 3, …

P35.82 The person and lightbulb are separated by a distance d, and the light travels at a distance $3d$. This gives a maximum ratio of 3.00, and we see that a ratio of 3.10 is impossible.

P35.84 (a–c) See P35.84 for full explanations.

P35.86 (a) $R_3 = nR_1$; (b) $R_3 = R_2$

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